COMP3153/9153

Algorithmic Verification

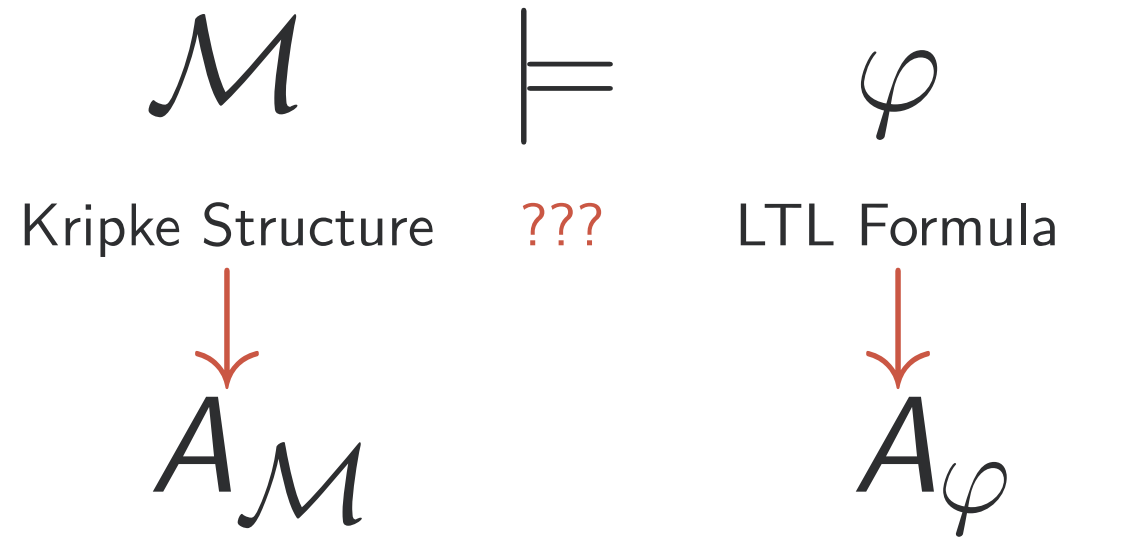
Lecture 7: LTL Model Checking and Büchi Automata

LTL Model Checking

$$\mathcal{M} \quad \models \quad \varphi$$

Kripke Structure ??? LTL Formula

LTL Model Checking



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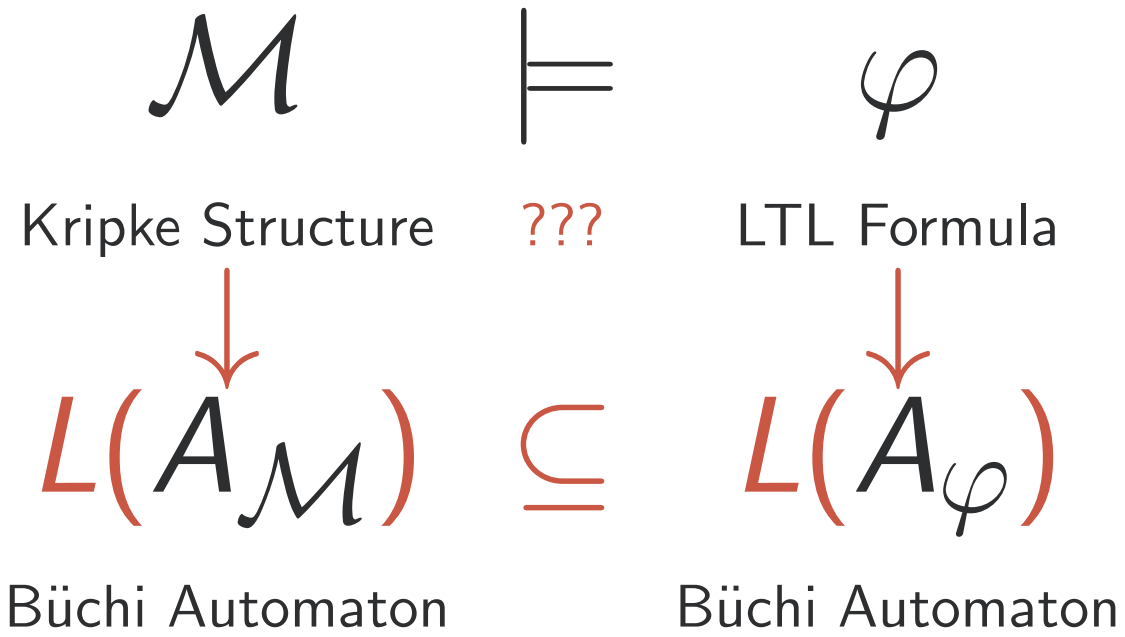

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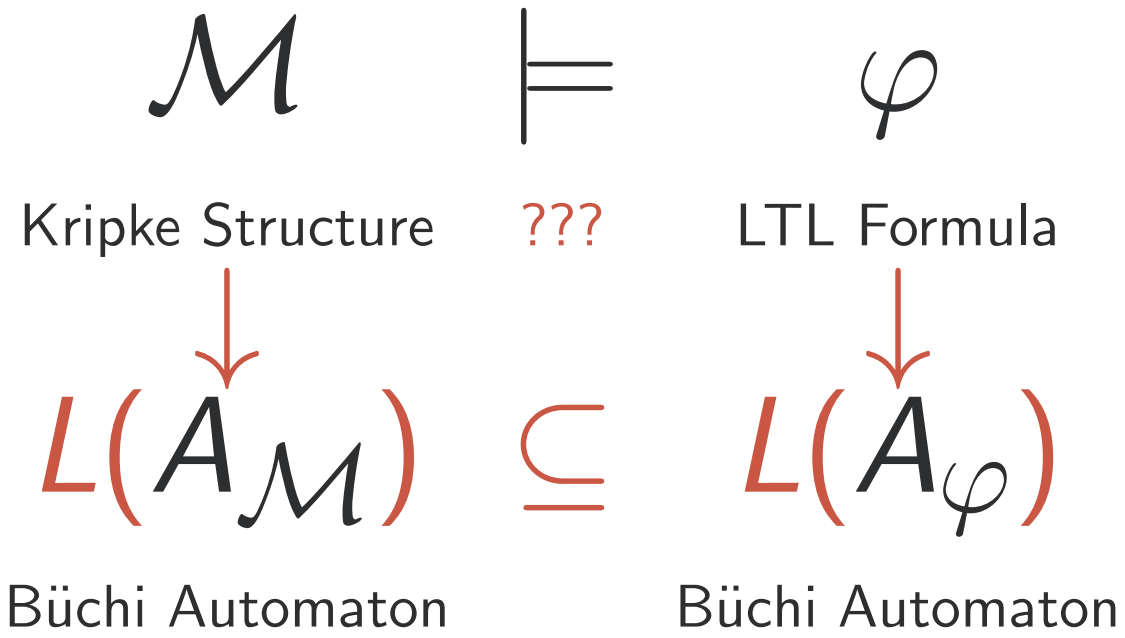

Büchi Automata

Büchi Automata are like finite automata, but their languages are of infinite-length strings, so they work well for behaviours $\in (2^{\mathcal{P}})^{\omega}$.

Büchi Automata

Definition

A (generalized) Büchi automaton is a 5-tuple $(Q, I, \Sigma, \delta, F)$ where

- Q is a set of states.
- $I \subseteq Q$ is a set of initial states.
- Σ is our alphabet of actions.
- $\delta : (Q \times \Sigma) \to 2^Q$ is our transition relation.
- $F \subseteq Q$ is a set of final states.

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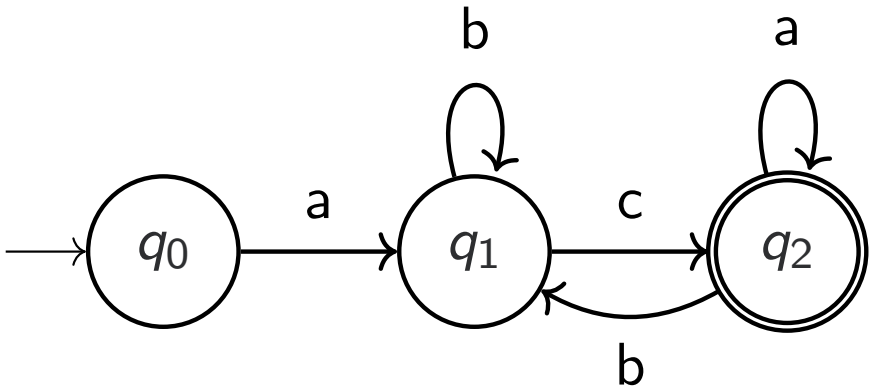
Language

We consider $\sigma \in L(A)$ for a Büchi automaton A iff it visits a particular final state infinitely often. More formally, define $\inf(\rho) = \{\ q \mid q \text{ appears infinitely often in } \rho\ \}$, then we say

$$\text{trace}(\rho) \in L(A) \Leftrightarrow \inf(\rho) \cap F \neq \emptyset$$

Example



- acaaaaaaa...

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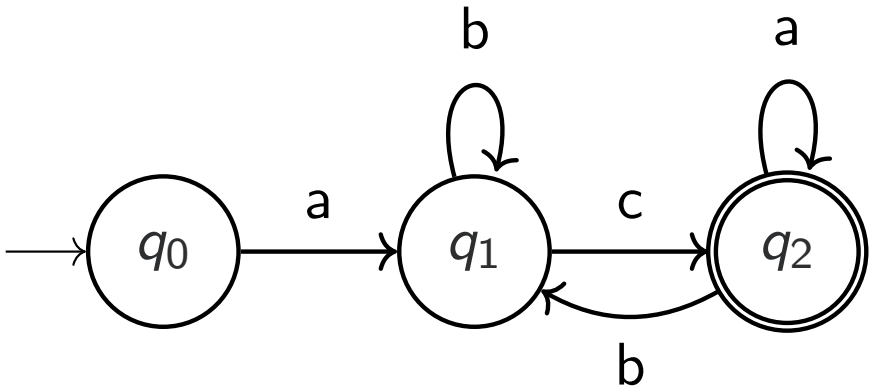

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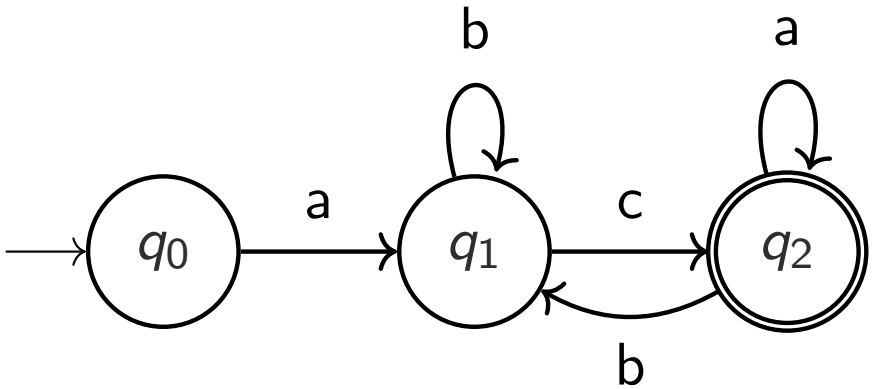

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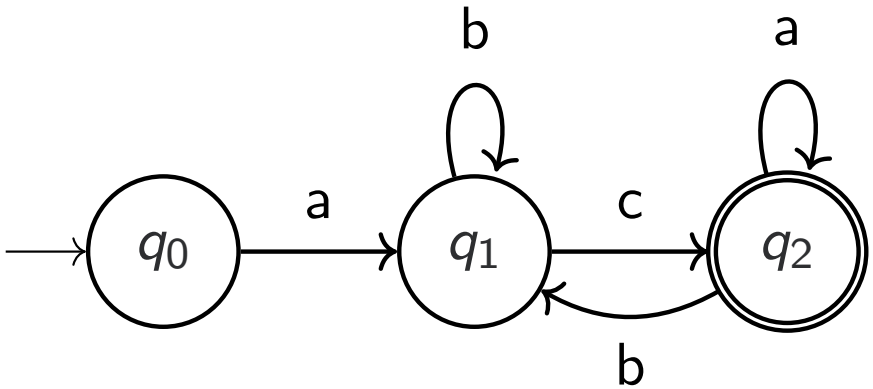

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Exercise

Let $\Sigma = \{0, 1\}$. Define Büchi automata for the following languages.

- $L_1 = \{v \in \Sigma^\omega \mid 0 \text{ occurs in } v \text{ exactly once } \}$

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- $L_4 = (01)^*\Sigma^\omega$

Closure Properties

Büchi Automata are closed under:

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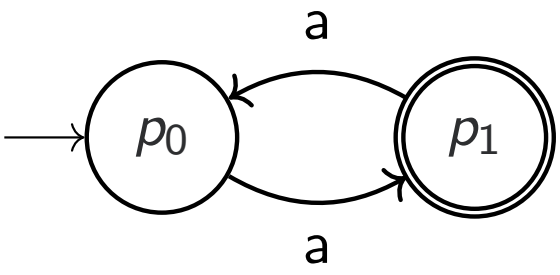
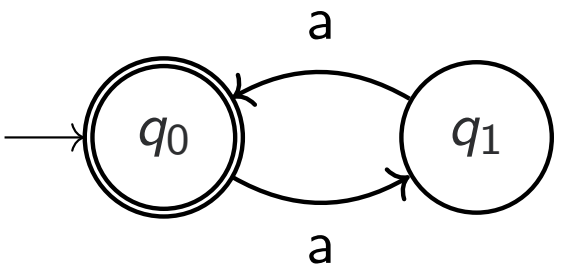
Intersection of GBAs





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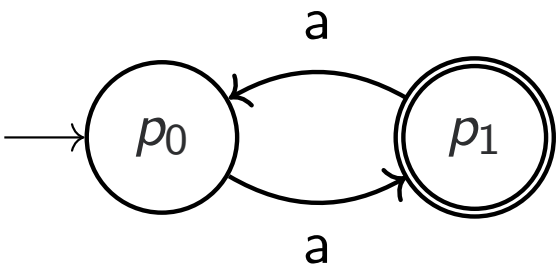

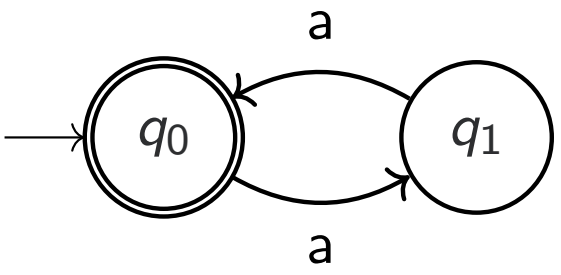

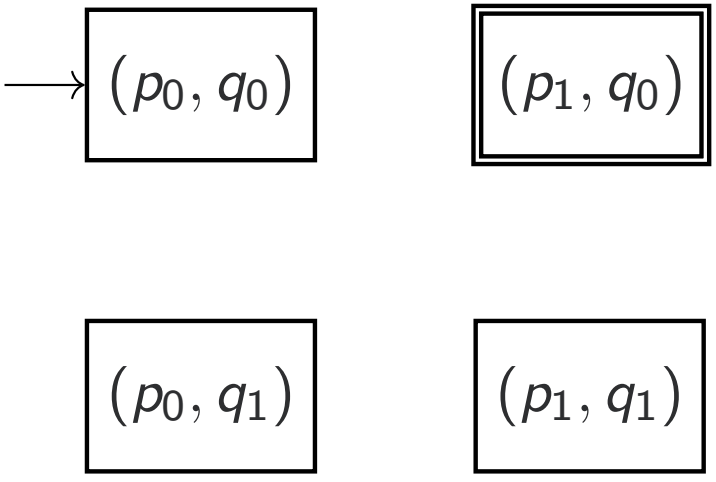

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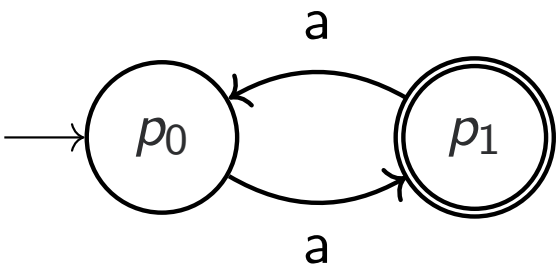

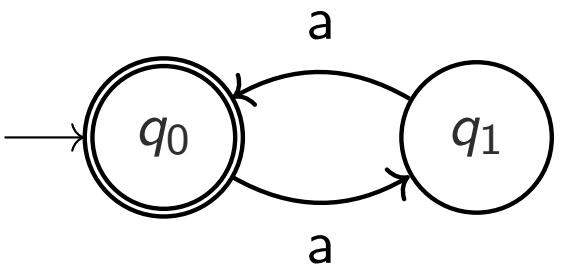

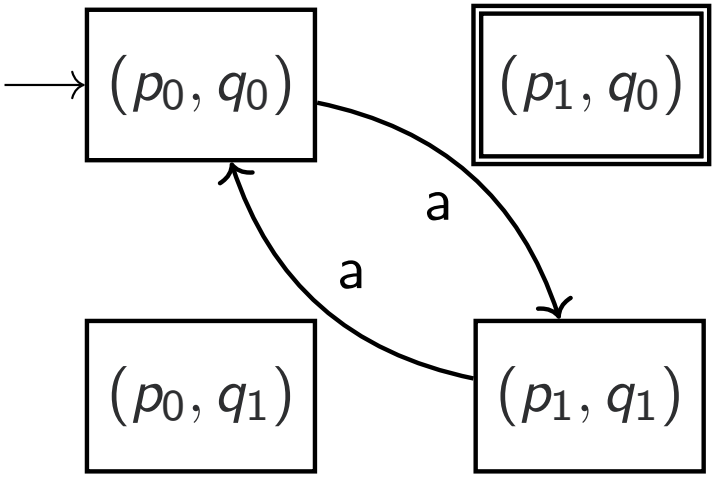

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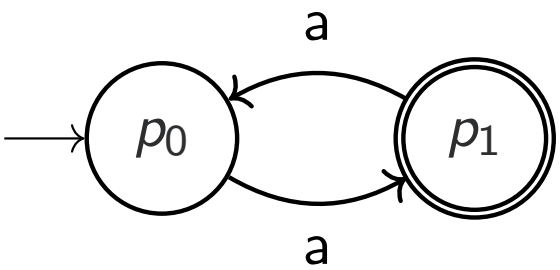

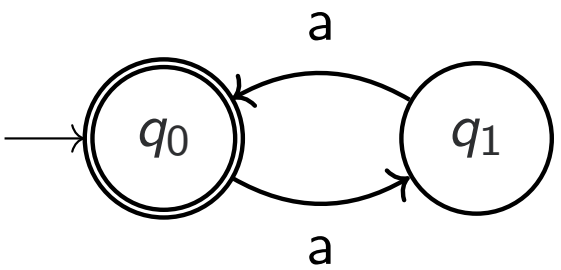

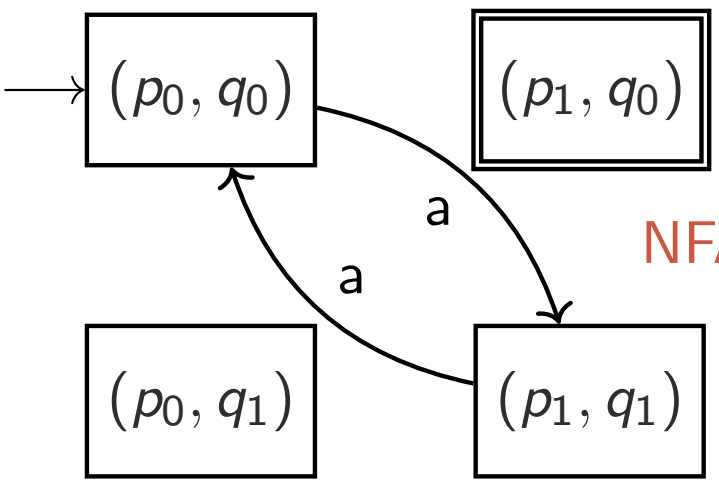

NFA product doesn't work!

Triple Product

An accepting cycle of a product of Büchi automata $P \times Q$ must cycle through accepting states of both P and Q infinitely often. Arbitrarily, we shall say it must alternate by visiting a final state of Q then P then Q and so on. This doesn't affect expressivity because we are only concerned with infinite strings.

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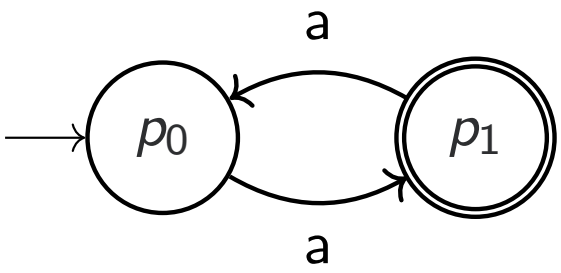
Key idea

Make three copies of the product: $P \times Q \times \{0, 1, 2\}$.

- Copy '0' is marked with initial states $I_P \times I_Q$.
- Copy '2' is entirely marked as final states.
- Transition relation like normal product, but:
  - We move from copy 0 to copy 1 when moving to a state $\in F_Q$.
  - We move from copy 1 to copy 2 when moving to a state $\in F_P$.
  - All transitions from copy 2 move back to copy 0.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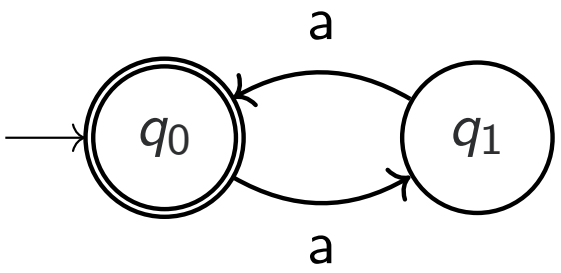
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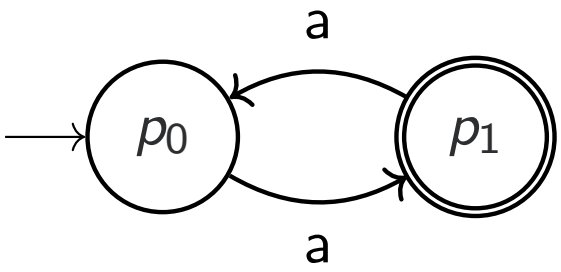
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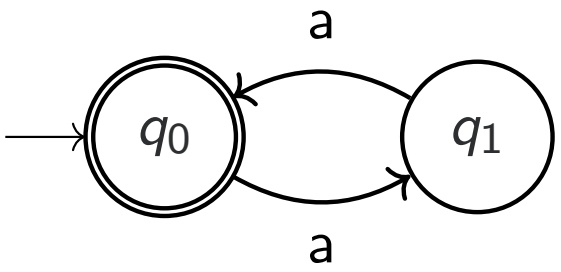
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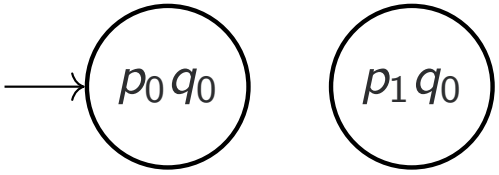
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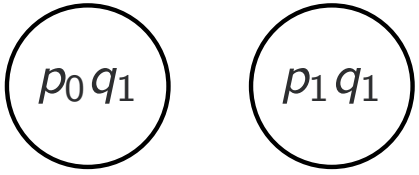
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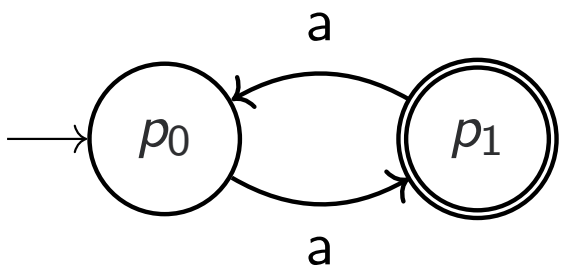
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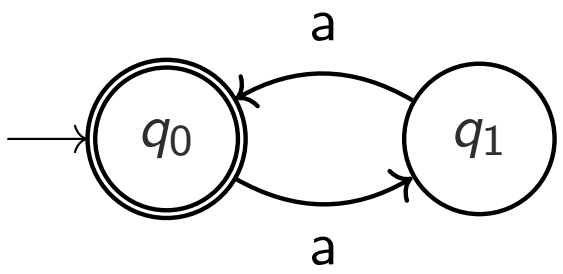
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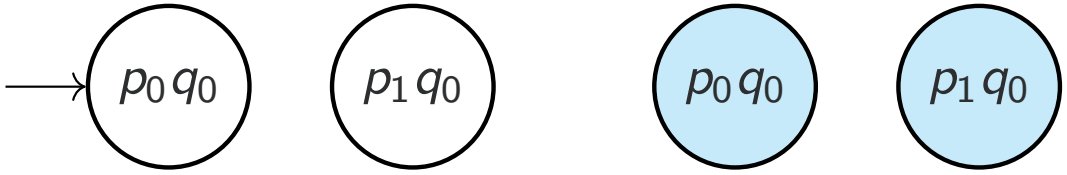
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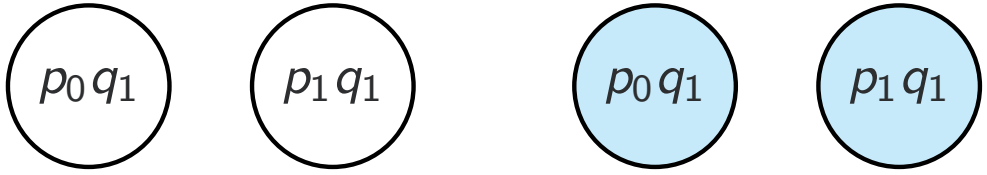
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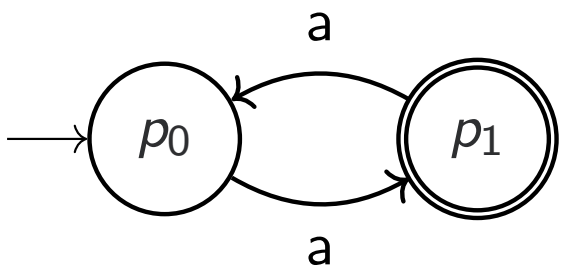
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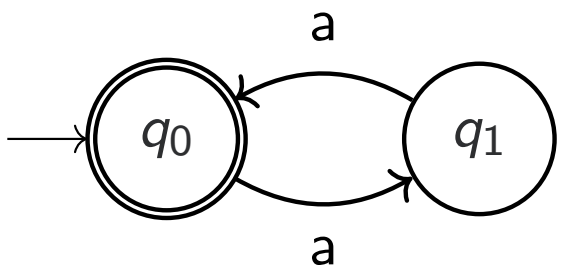
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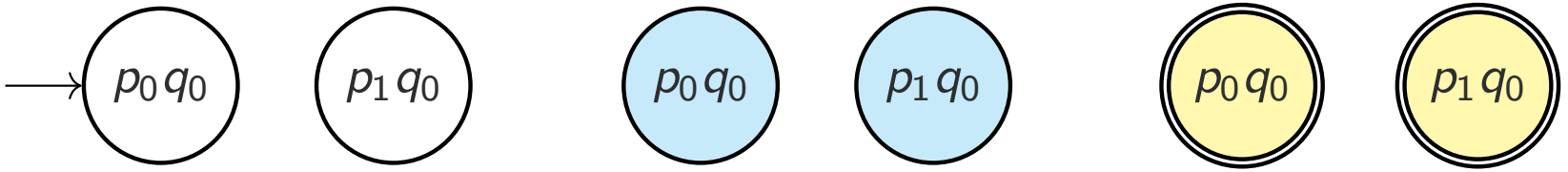
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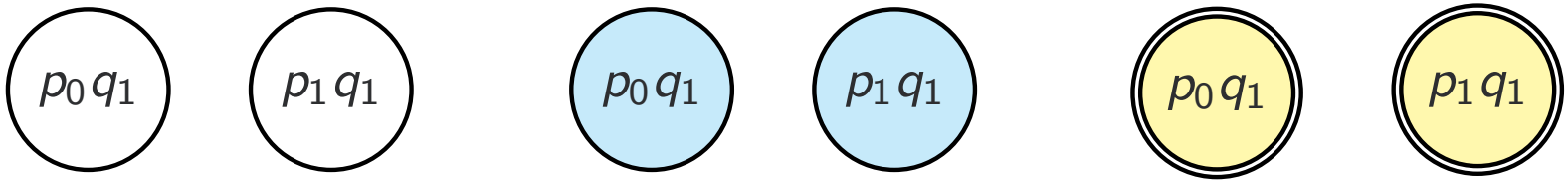
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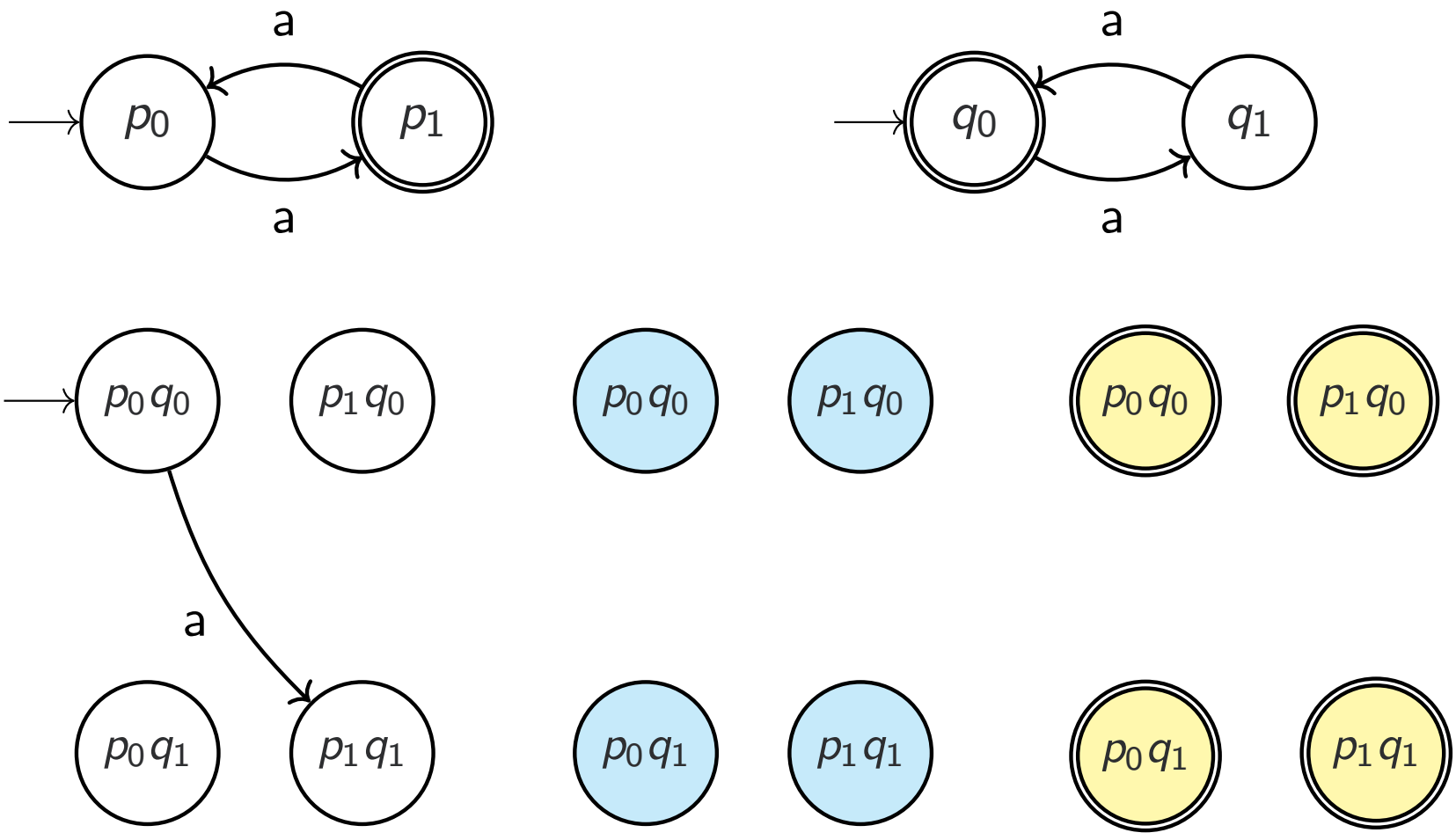
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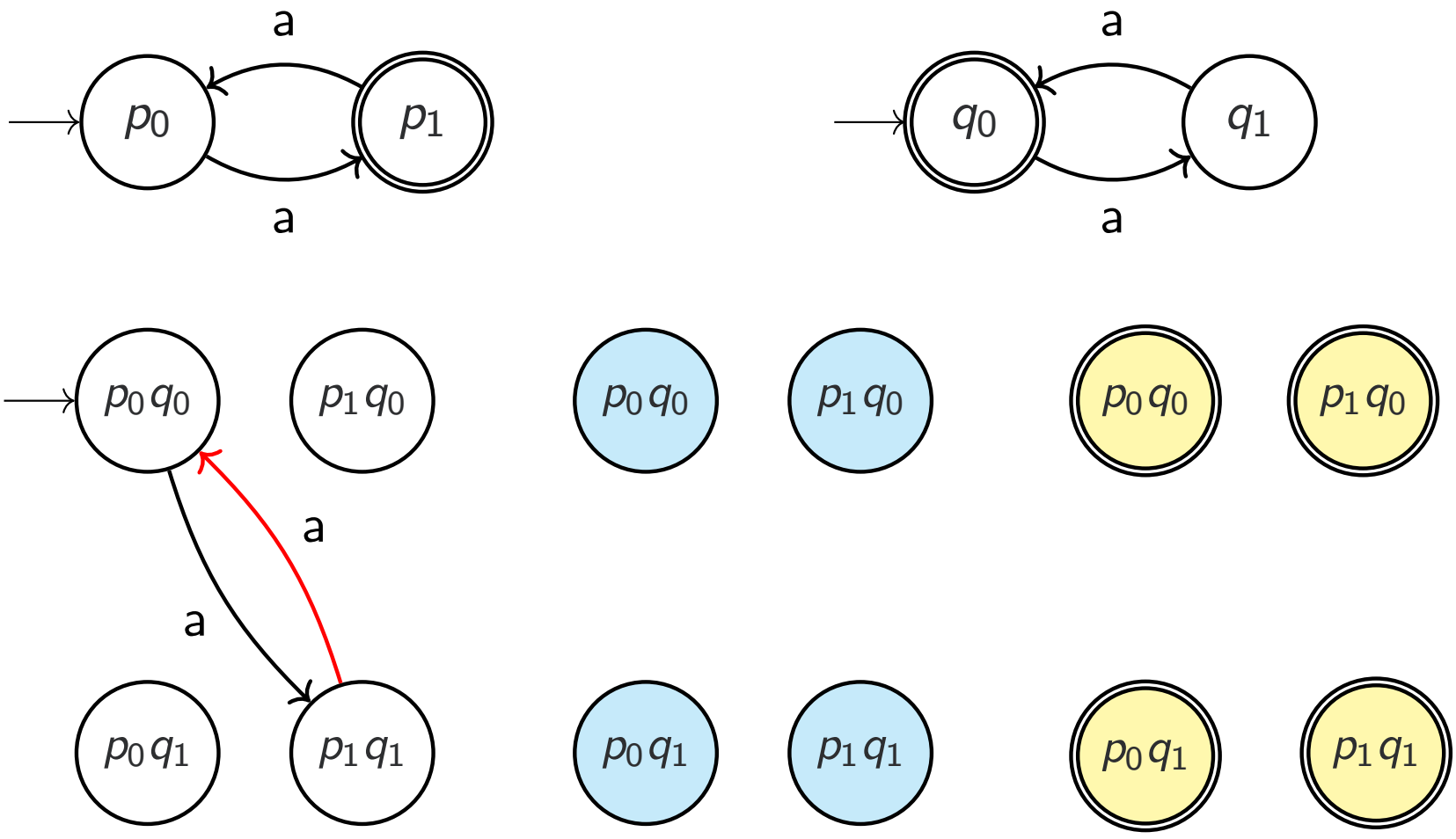
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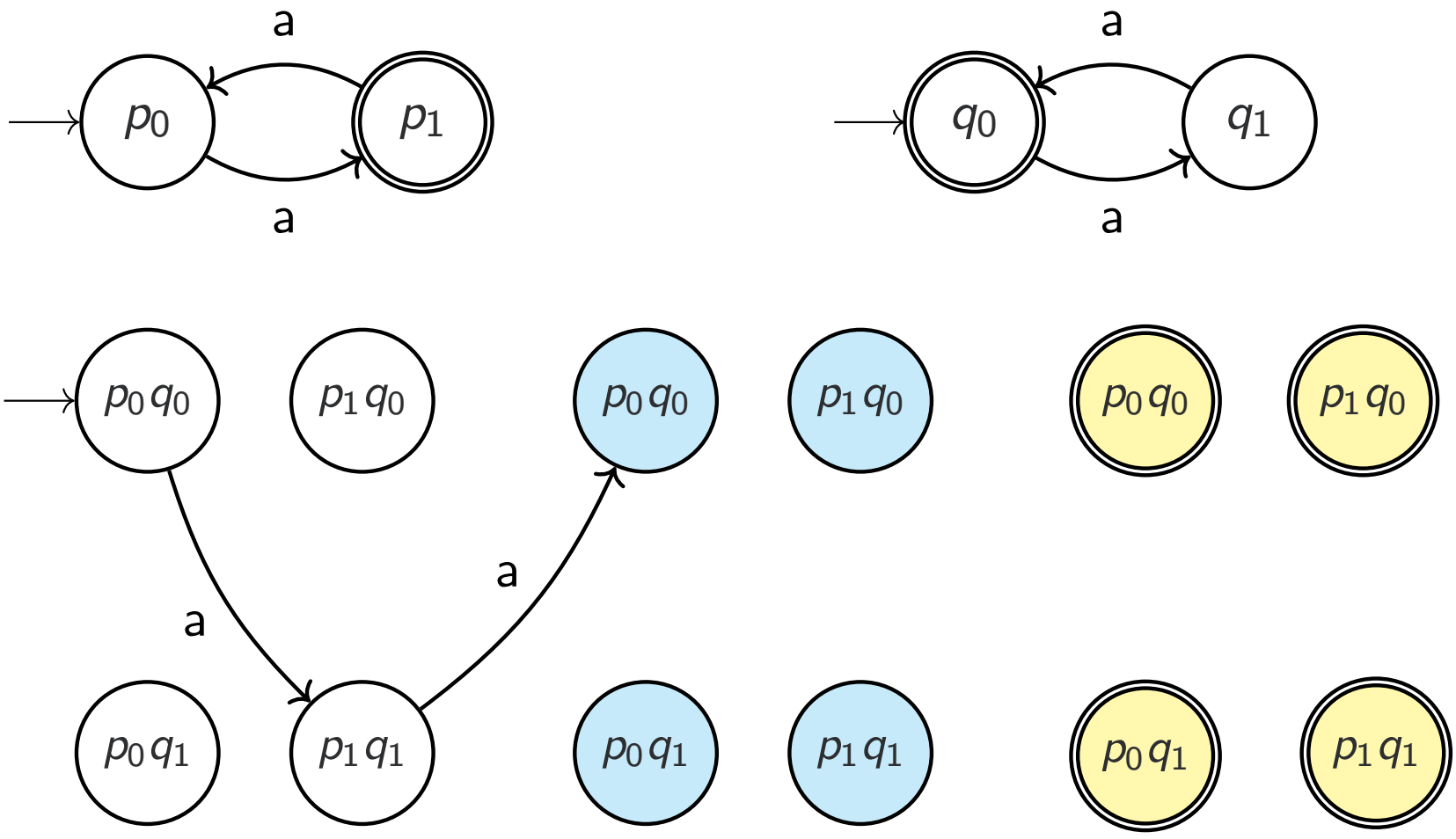
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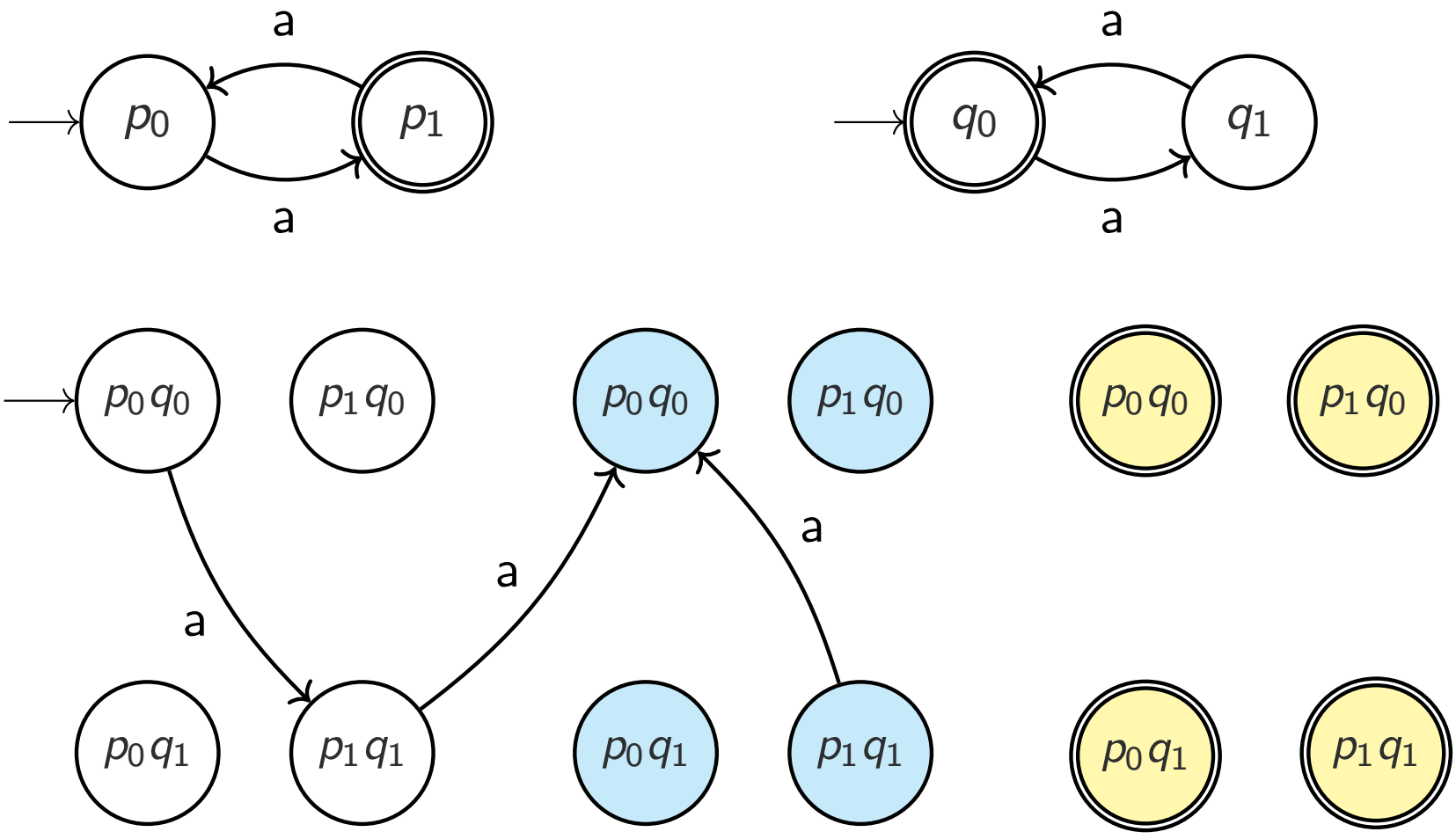
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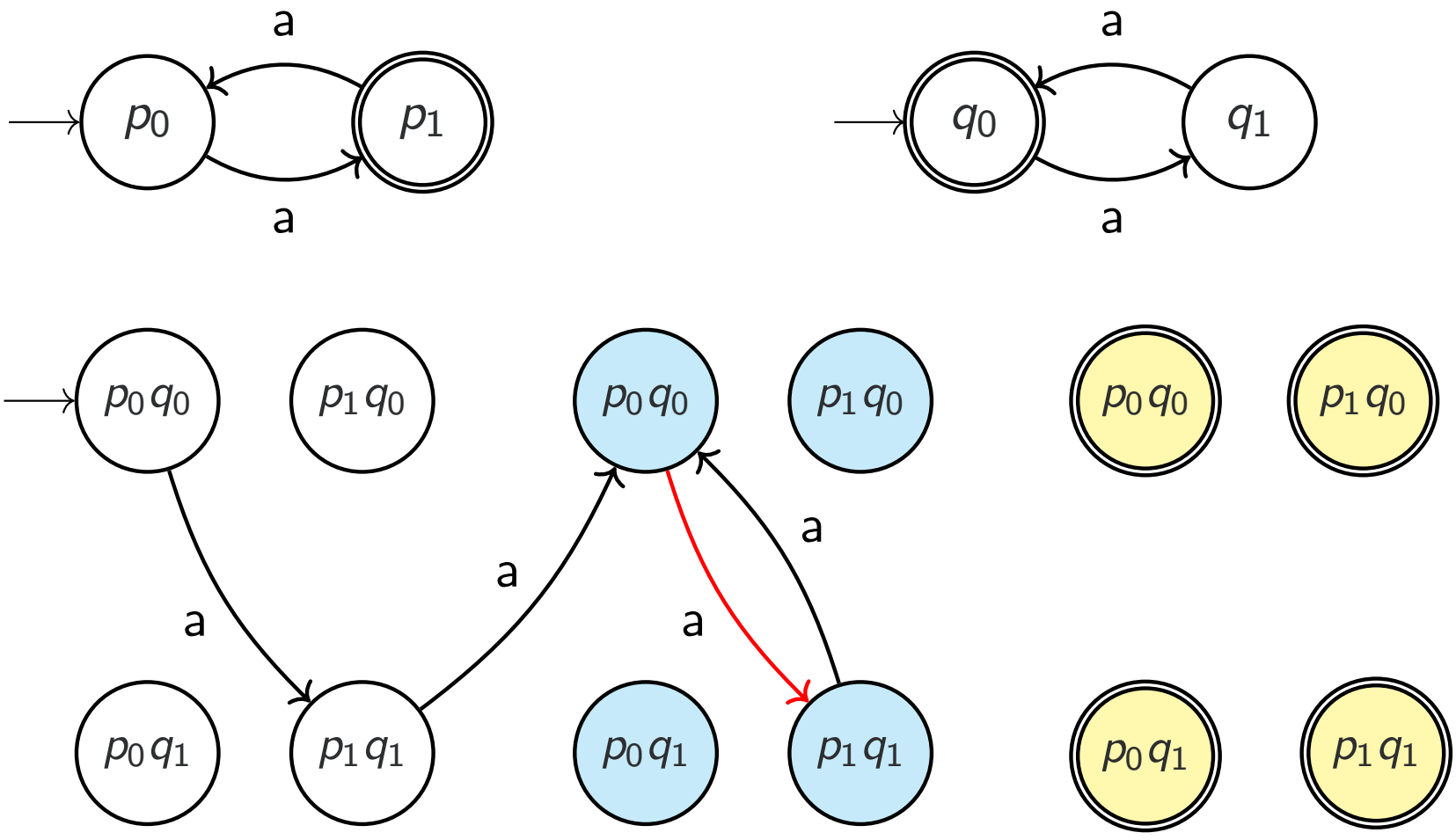
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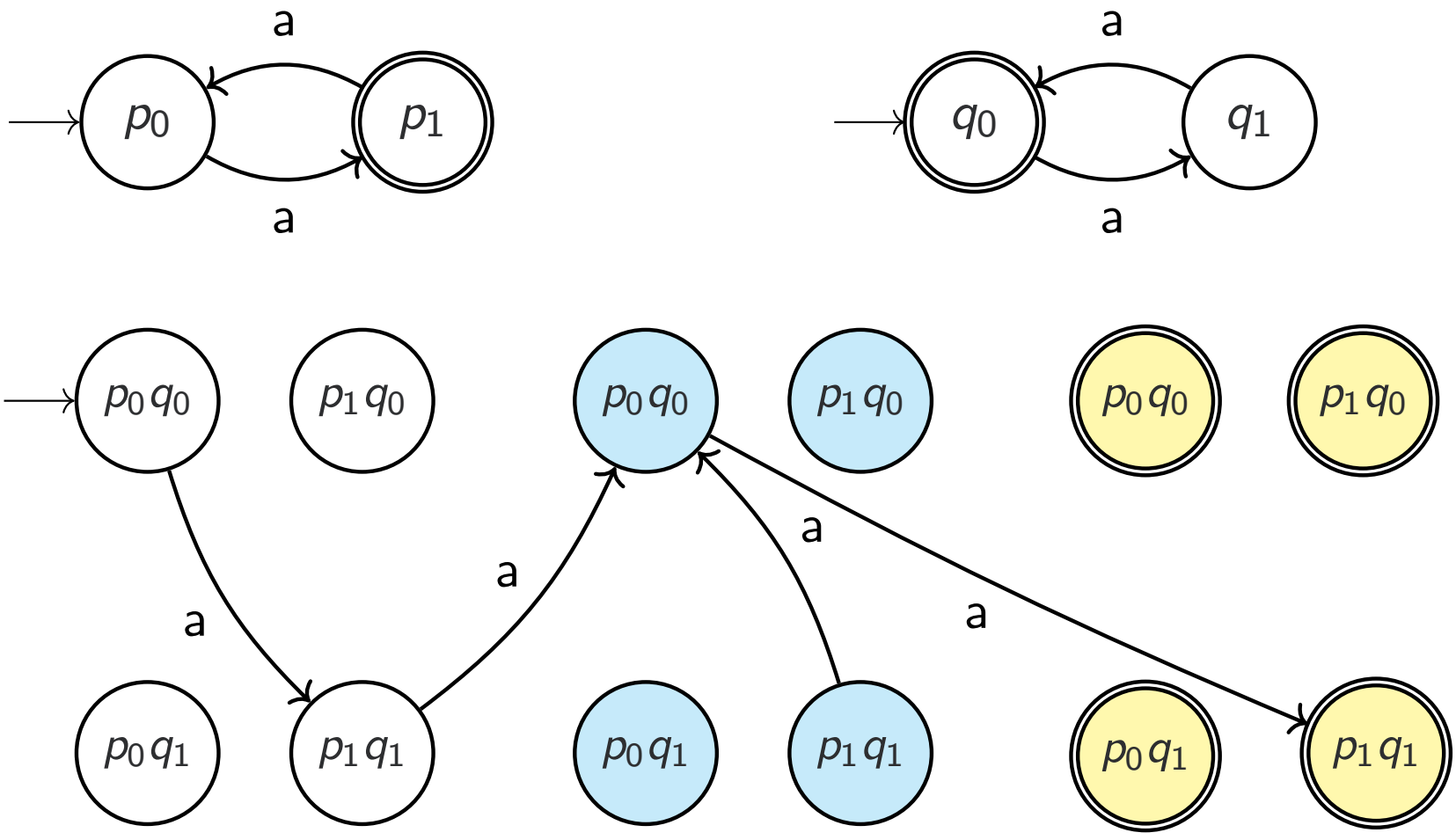
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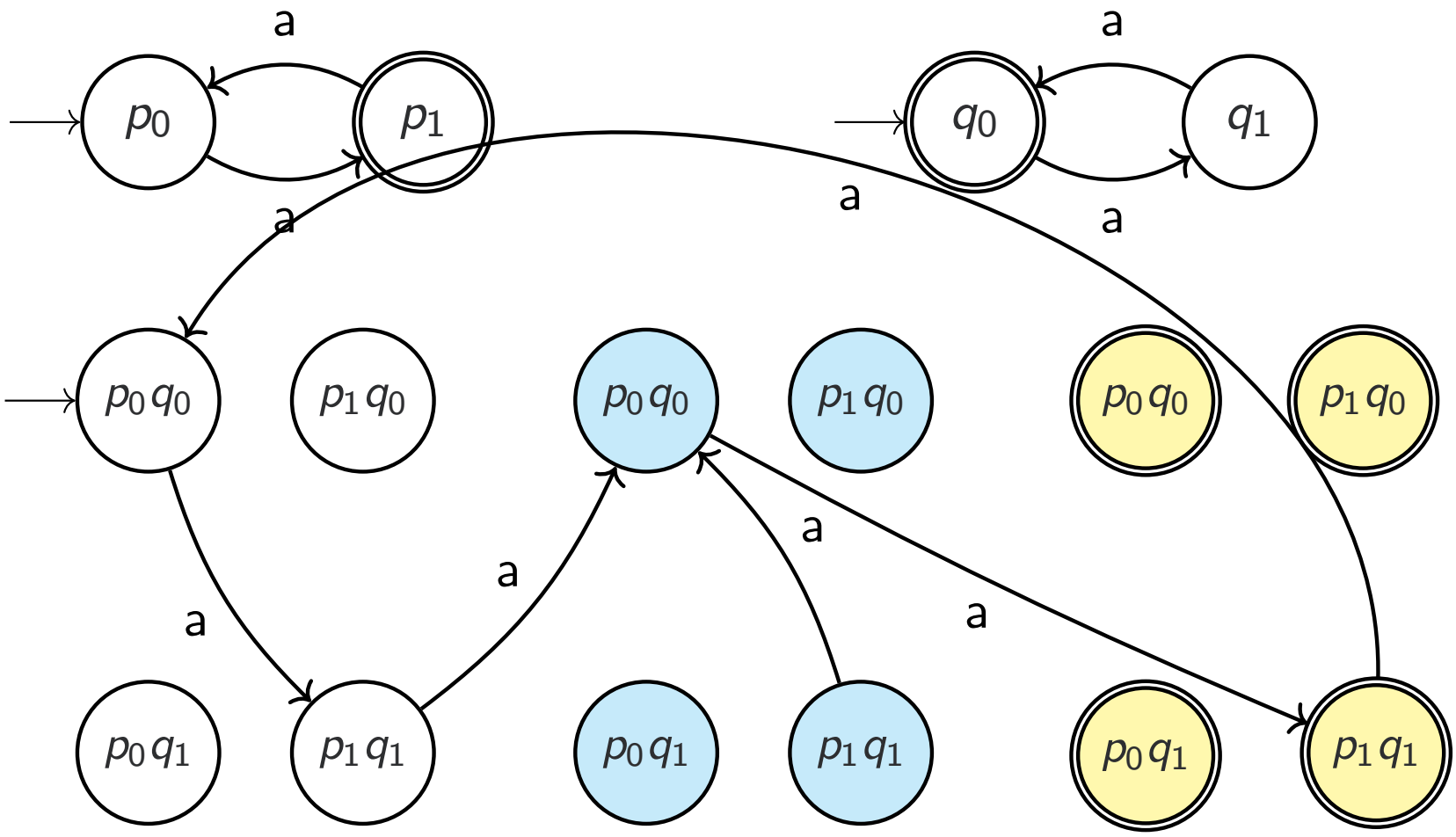
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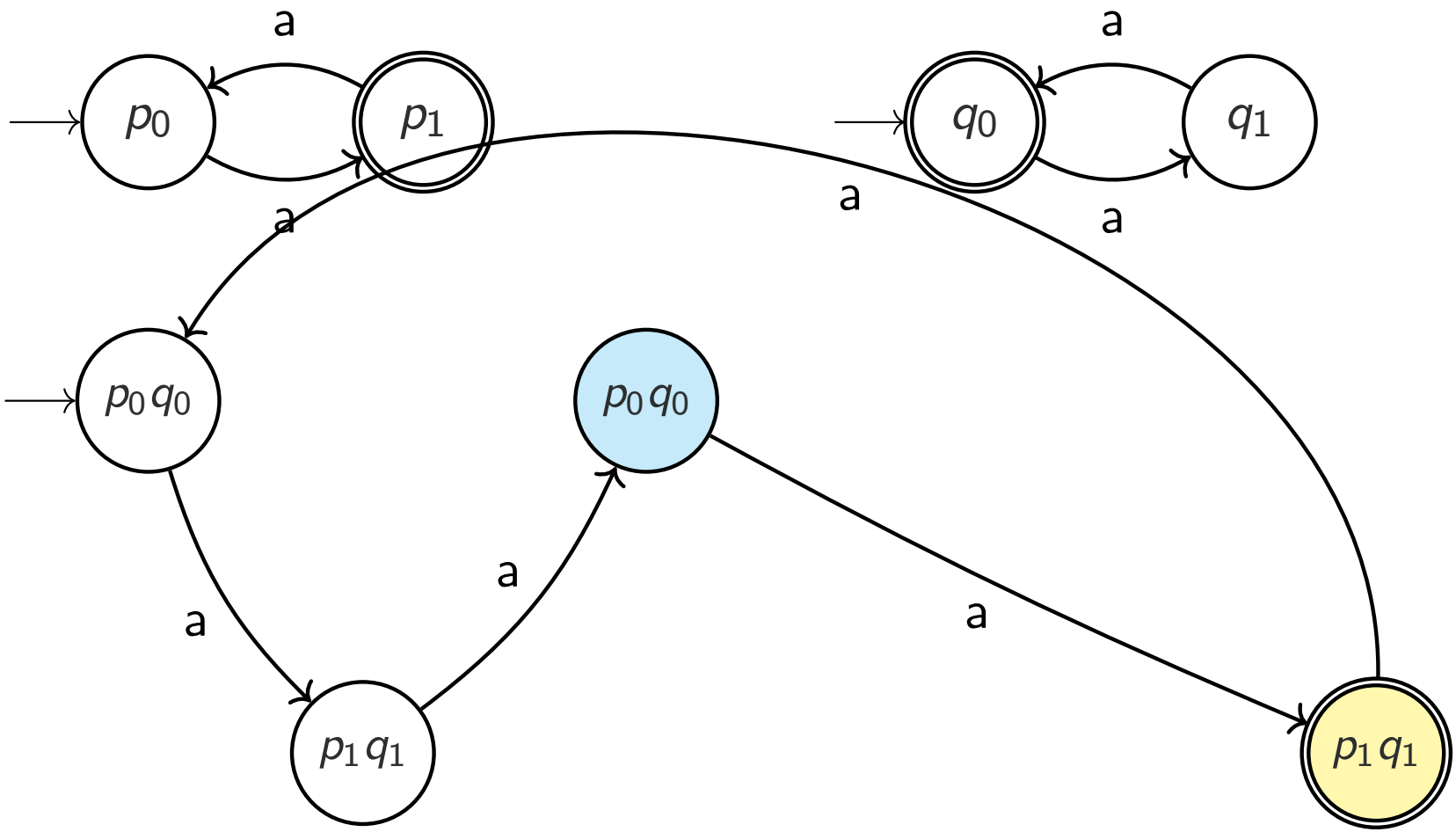
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Büchi Product

Let $A_1 = (Q_1, I_1, \Sigma_1, \delta_1, F_1)$ and $A_2 = (Q_2, I_2, \Sigma_2, \delta_2, F_2)$.

Definition

Define $A_\cap$ with $Q = Q_1 \times Q_2 \times \{0, 1, 2\}$, and $I = I_1 \times I_2 \times \{0\}$, $\Sigma = \Sigma_1 \cap \Sigma_2$ and $F = Q_1 \times Q_2 \times \{3\}$.
We define δ as follows:

$$
\begin{array}{lll}
((q_1, q_2, 0), a, (q_1', q_2', 0)) \in \delta & \text{iff} & (q_i, a, q_i') \in \delta_i \ (i = 1, 2) \wedge q_1' \notin F_1 \\
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$$L(A_M) \subseteq L(A_\Phi)$$

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\equiv \quad & L(A_M) \cap L(A_\Phi)^C = \emptyset
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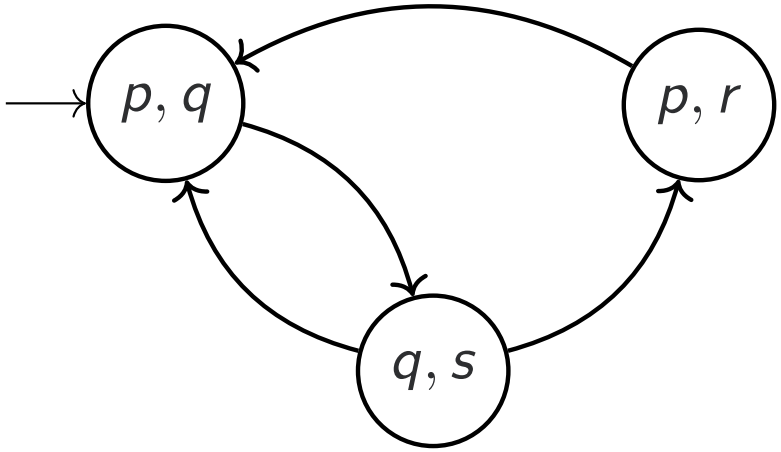
We still need to know how to:

- Determine if $L(A) = \emptyset$ for a Büchi automaton A.
- Convert a Kripke structure M to a Büchi automaton A_M
- Convert a LTL formula Φ to a Büchi automaton A_Φ.

Büchi from Kripke



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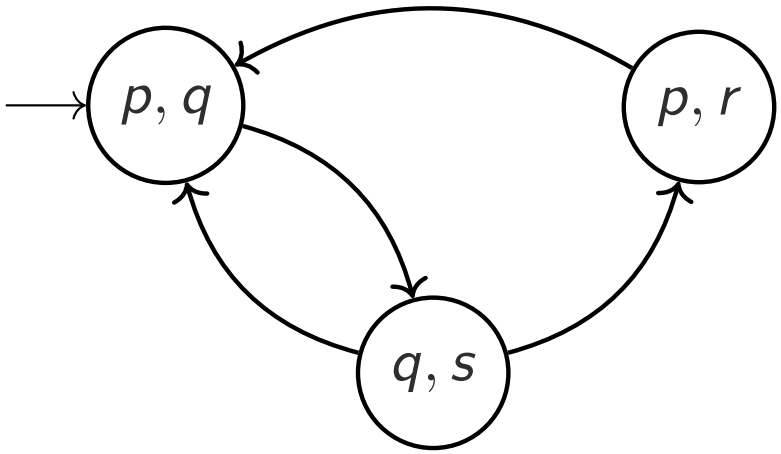
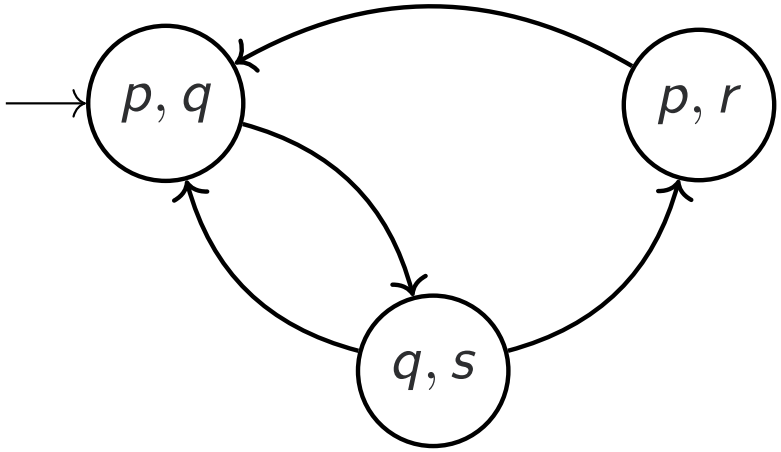


How to convert

We add a new initial state, move labels on the states to all incoming edges, and make all states final.

Büchi from Kripke



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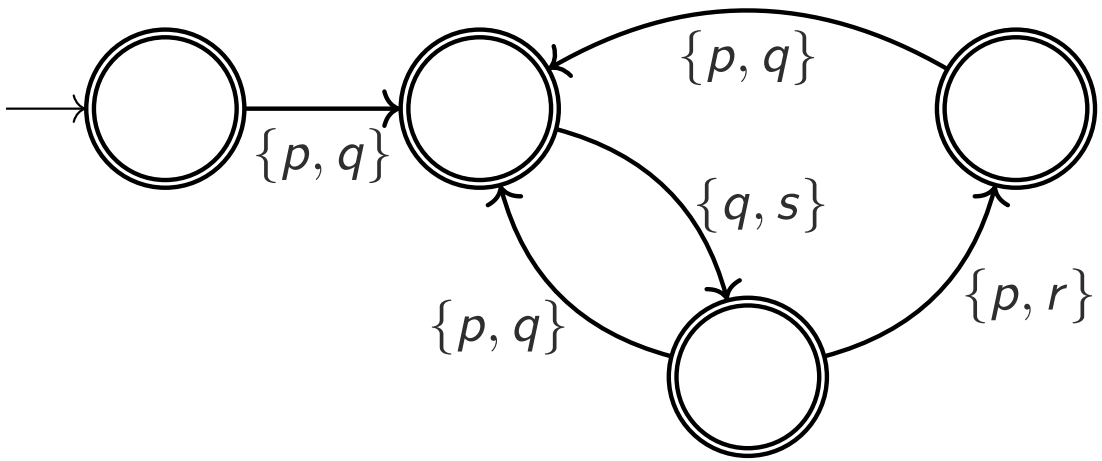

Büchi Automata Emptiness

Theorem (Büchi pumping)

Given a Büchi Automaton $A = (Q, I, \Sigma, \delta, F)$ then $L(A) \neq \emptyset$ iff there exists $v, w \in \Sigma^*$ with lengths $\leq |Q|$ such that $vw^\omega \in L(A)$.

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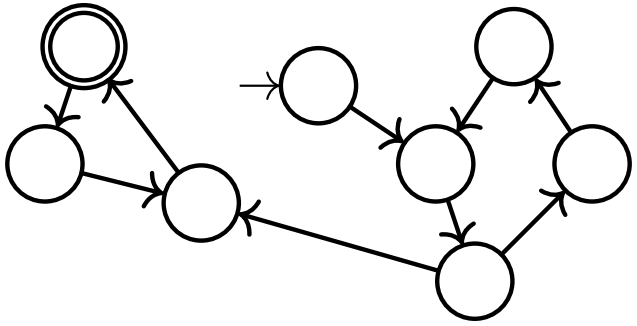

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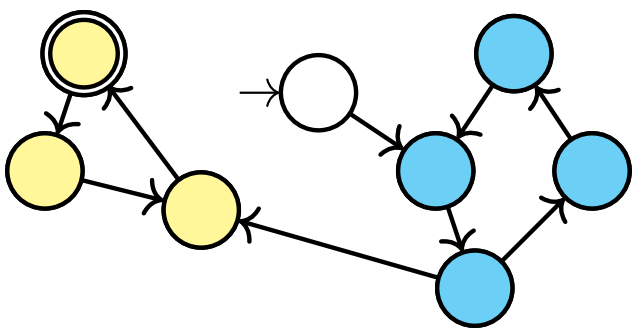
We use **Strongly Connected Components!**. Many algorithms exist (see online).



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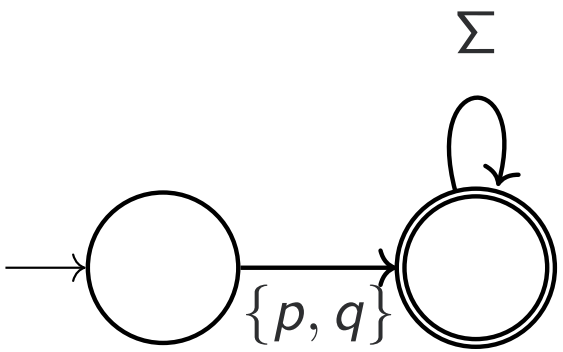

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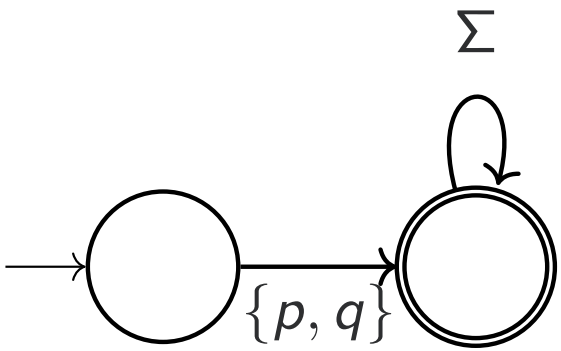

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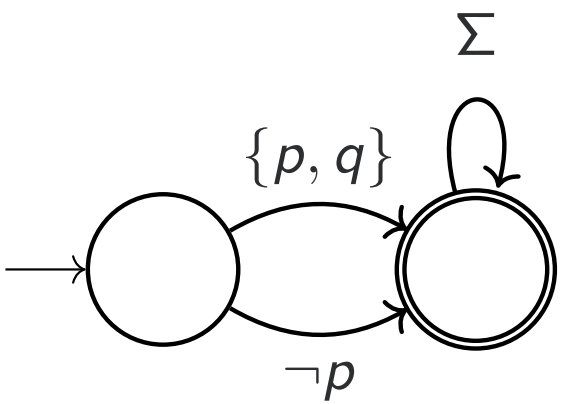

We can manually construct them for temporal formulae, but how to do so systematically?

Methods of LTL to Büchi

Many exist. All are complicated.

- Tableau Methods (Kersten, Manna, McGuire, Pnueli or Geth, Peled, Vardi, Wolper)
- Automata Theoretic (Vardi)
- Local and Eventuality Automata (Vardi, Wolper)

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4. Intersect the two automata, then reduce the alphabet to just atomic propositions.

Closure and Maximal Subsets

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The *closure* $\mathrm{Cl}(\Phi)$ of an LTL formula Φ is the set of all subformulae of Φ and their negation.

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Maximal Subsets

Define $\mathrm{Sub}(\Phi)$ of an LTL formula Φ as the set of all maximal subsets of $\mathrm{Cl}(\Phi)$ that are *locally consistent* (not contradictory).

Local Automaton

Definition

The local automaton for A^L_Φ for a formula Φ is defined as $(Q, I, \Sigma, \delta, F)$ where:

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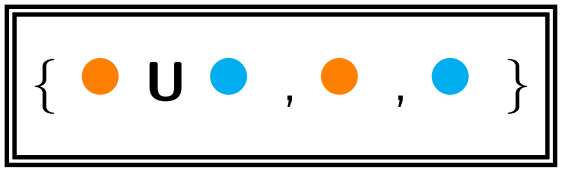
- $q \in \delta(p, a)$ if $a = p$ and
  - $\mathbf{X}\varphi \in p$ iff $\varphi \in q$
  - $\varphi\ \mathbf{U}\ \psi \in p$ iff $\psi \in p$ or $\varphi \in p \wedge (\varphi\ \mathbf{U}\ \psi) \in q$

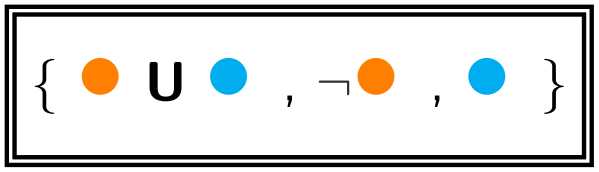
Whats the local automaton for $\mathbf{X}\bullet$?

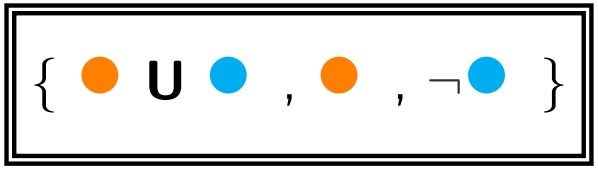
Example

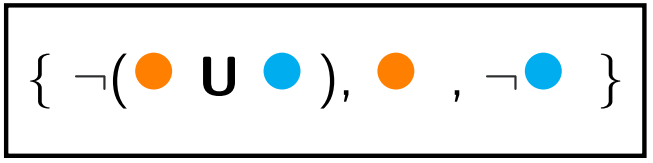
Whats the local automaton for ● **U** ● ?

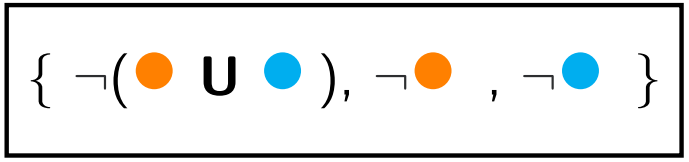
(the edge actions are always just the origin state, so they're omitted)











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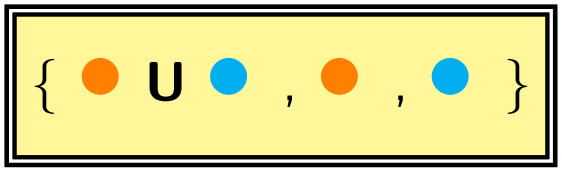

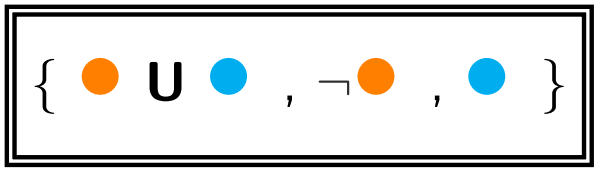

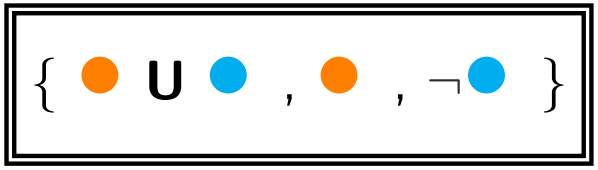

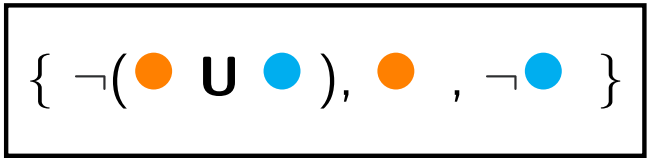

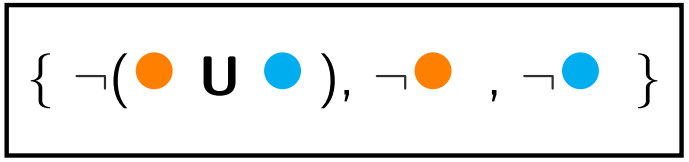

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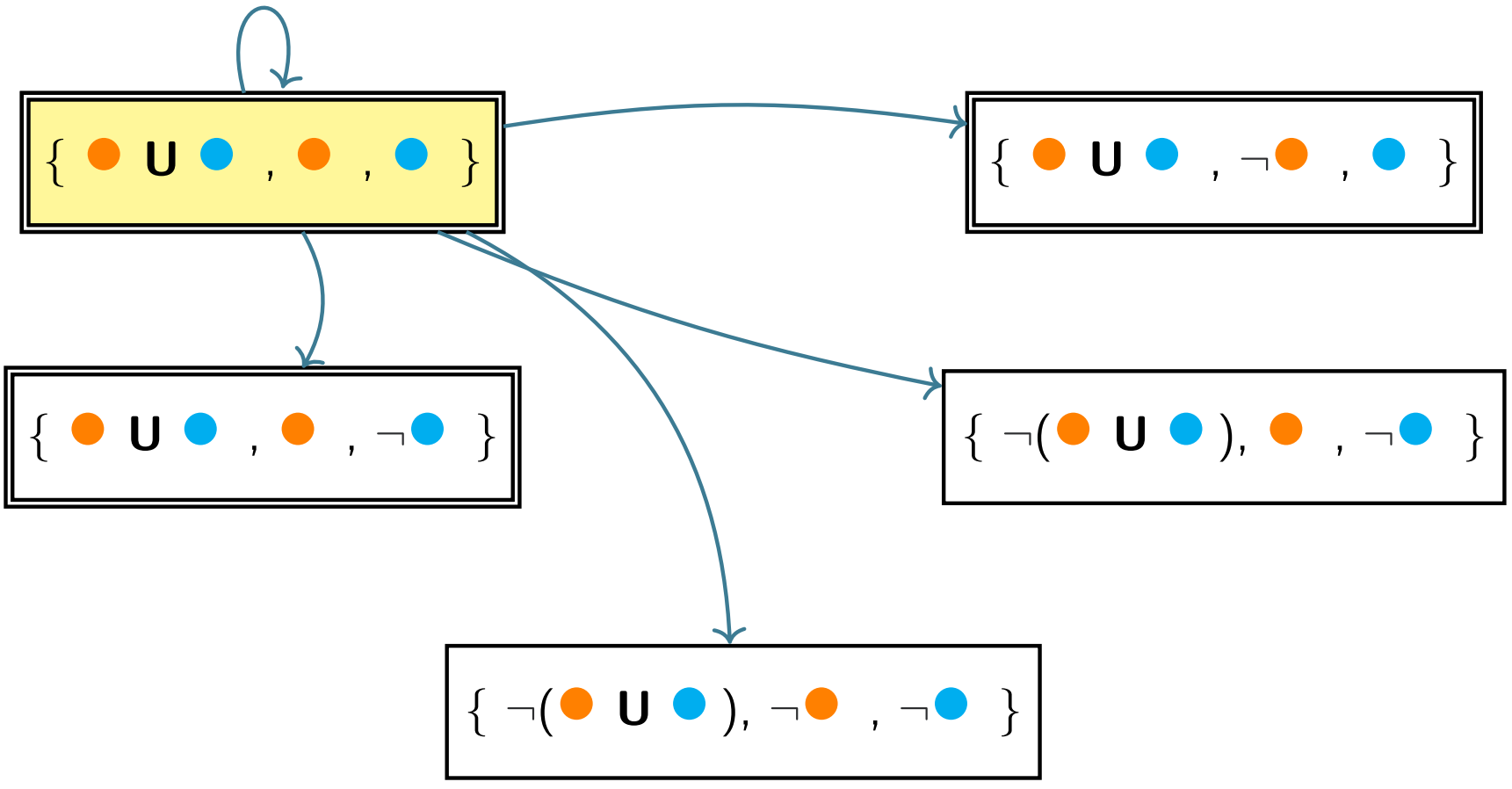

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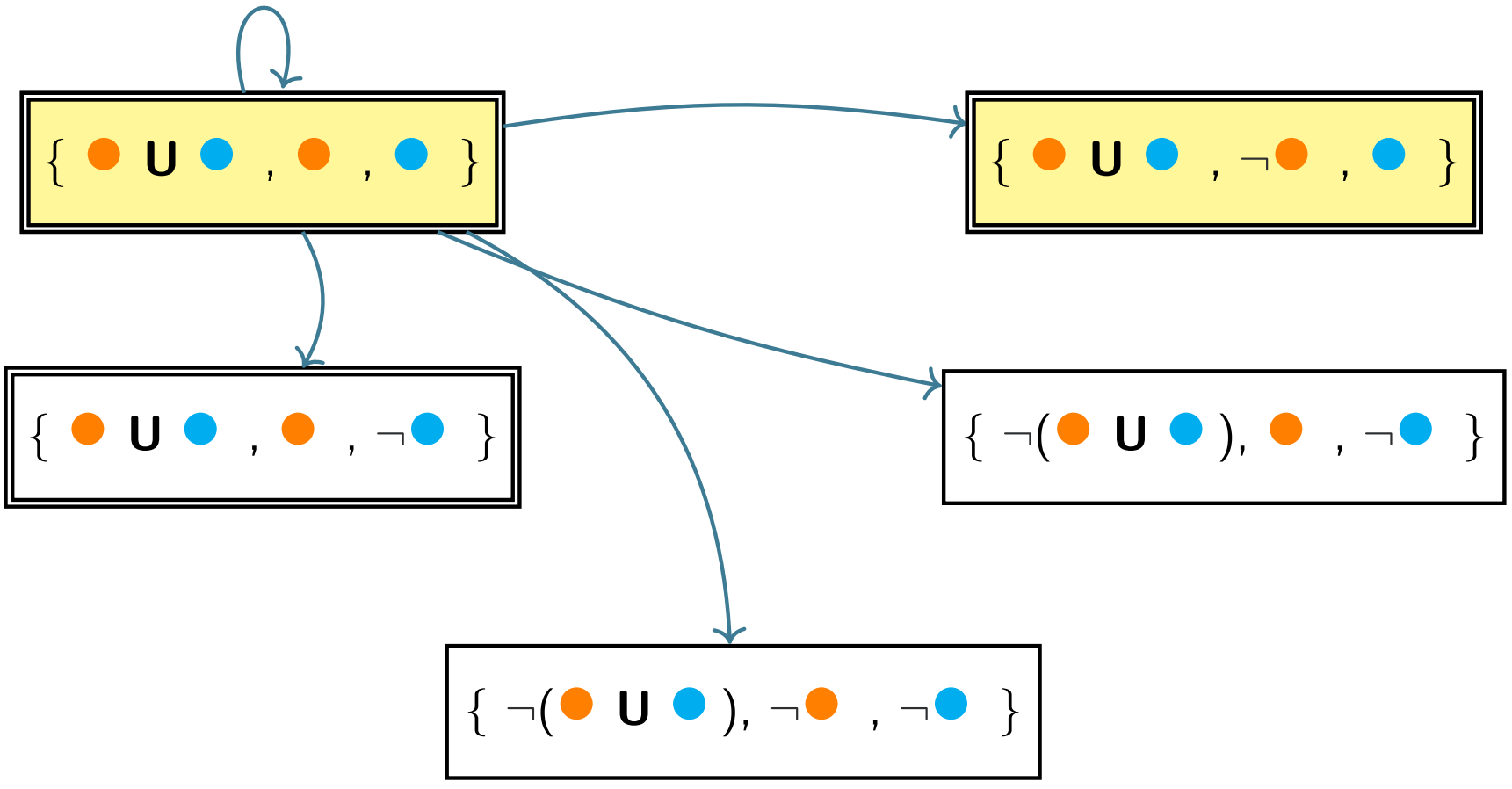

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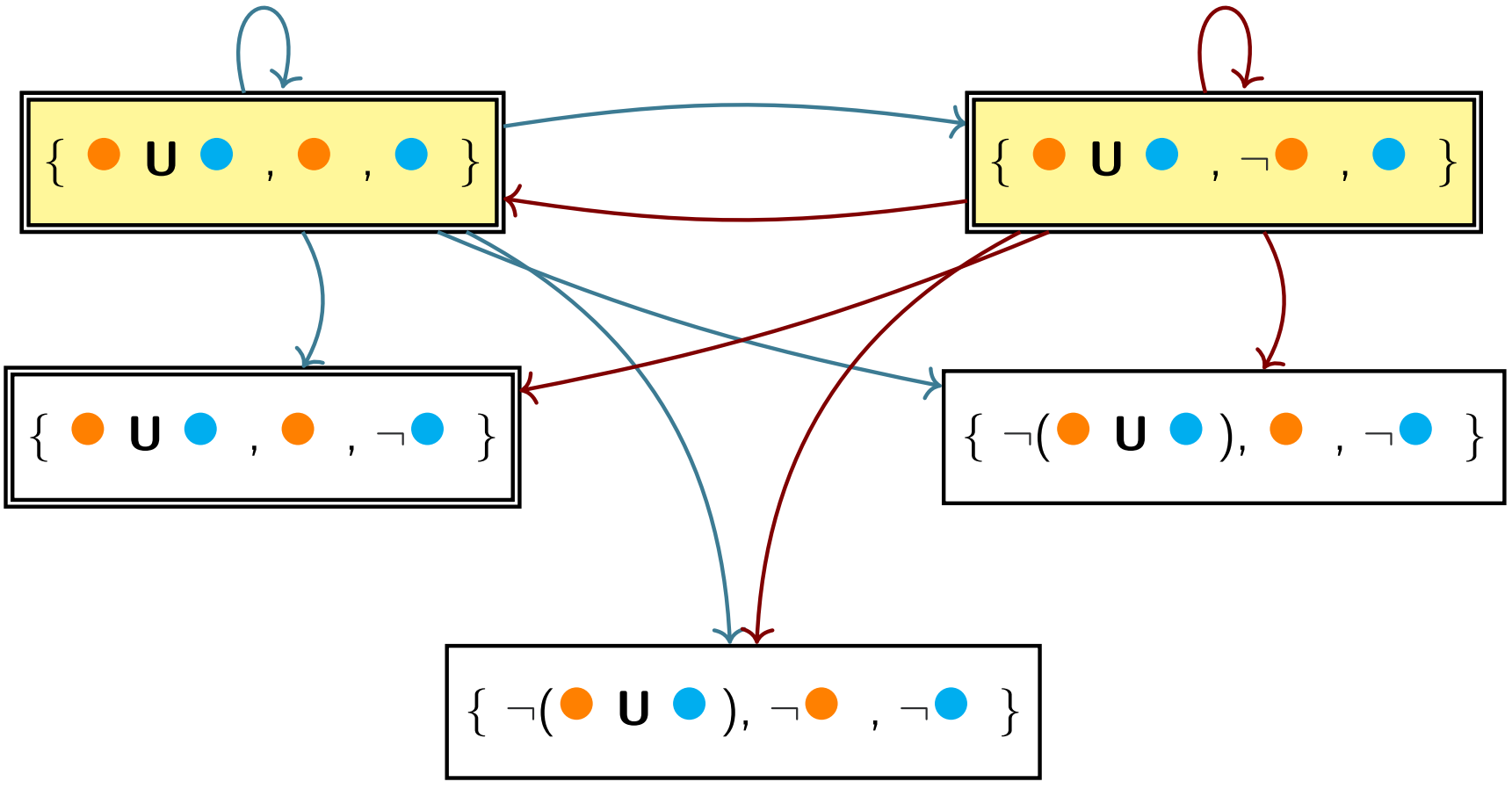

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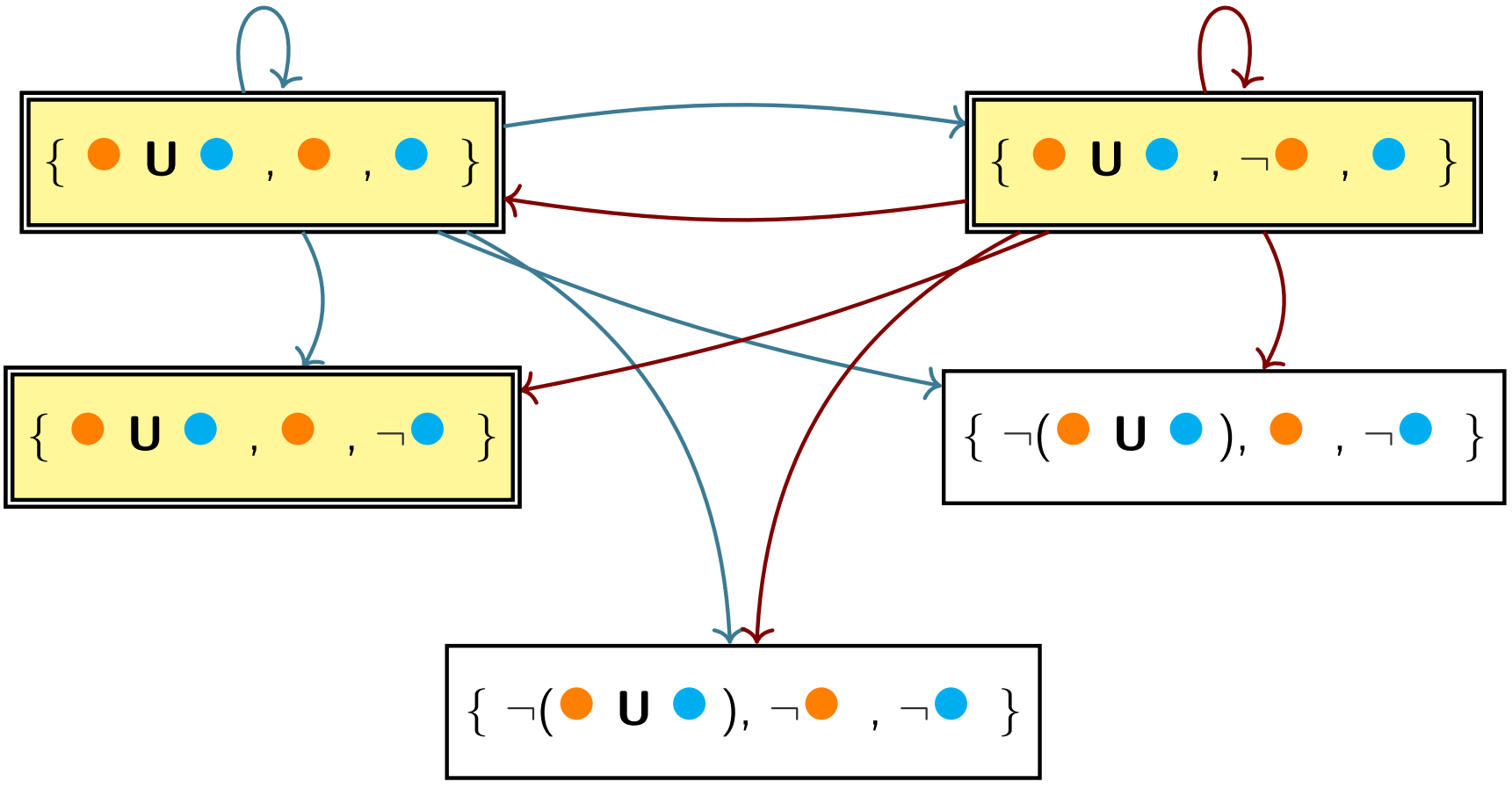

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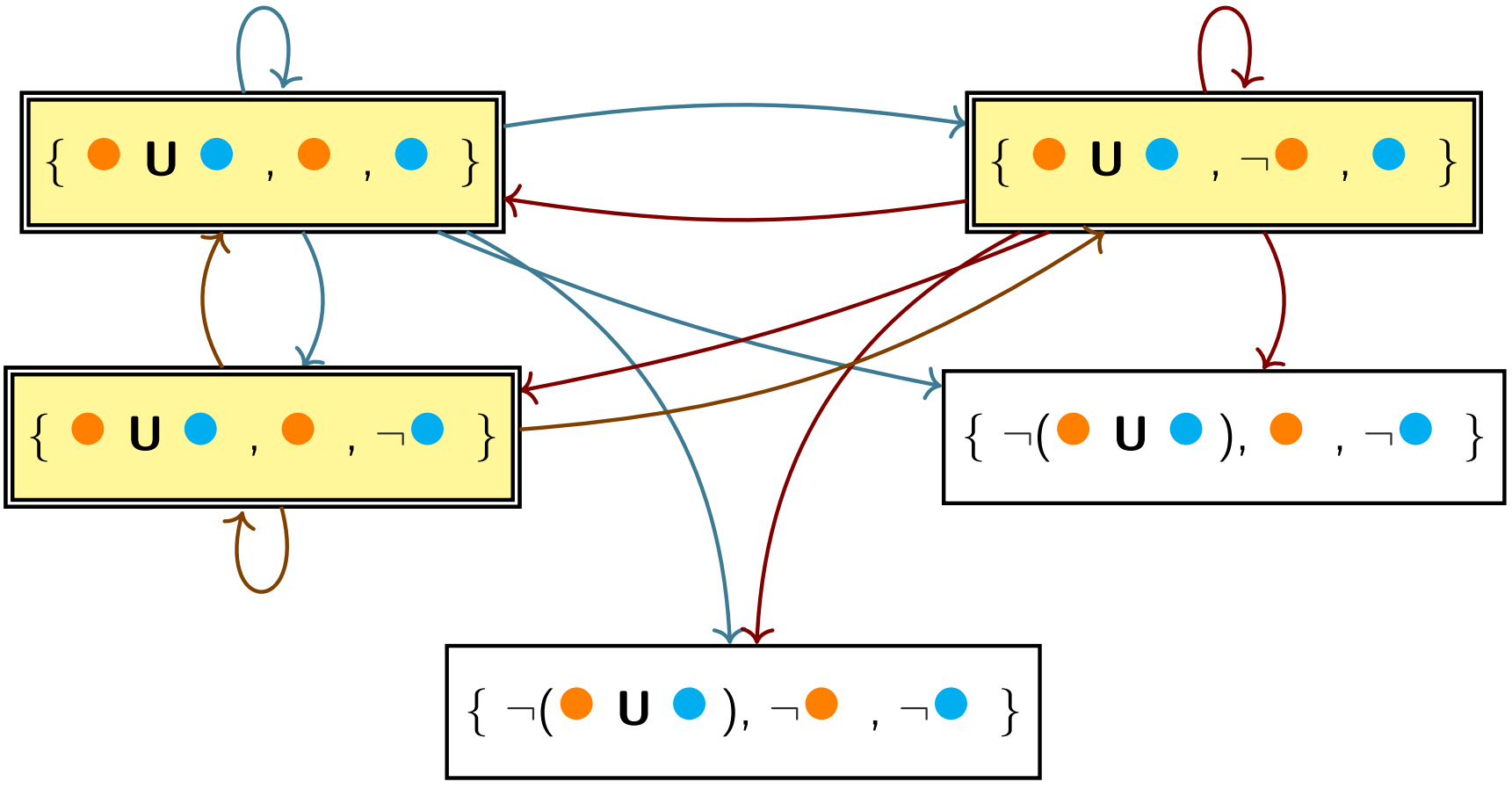

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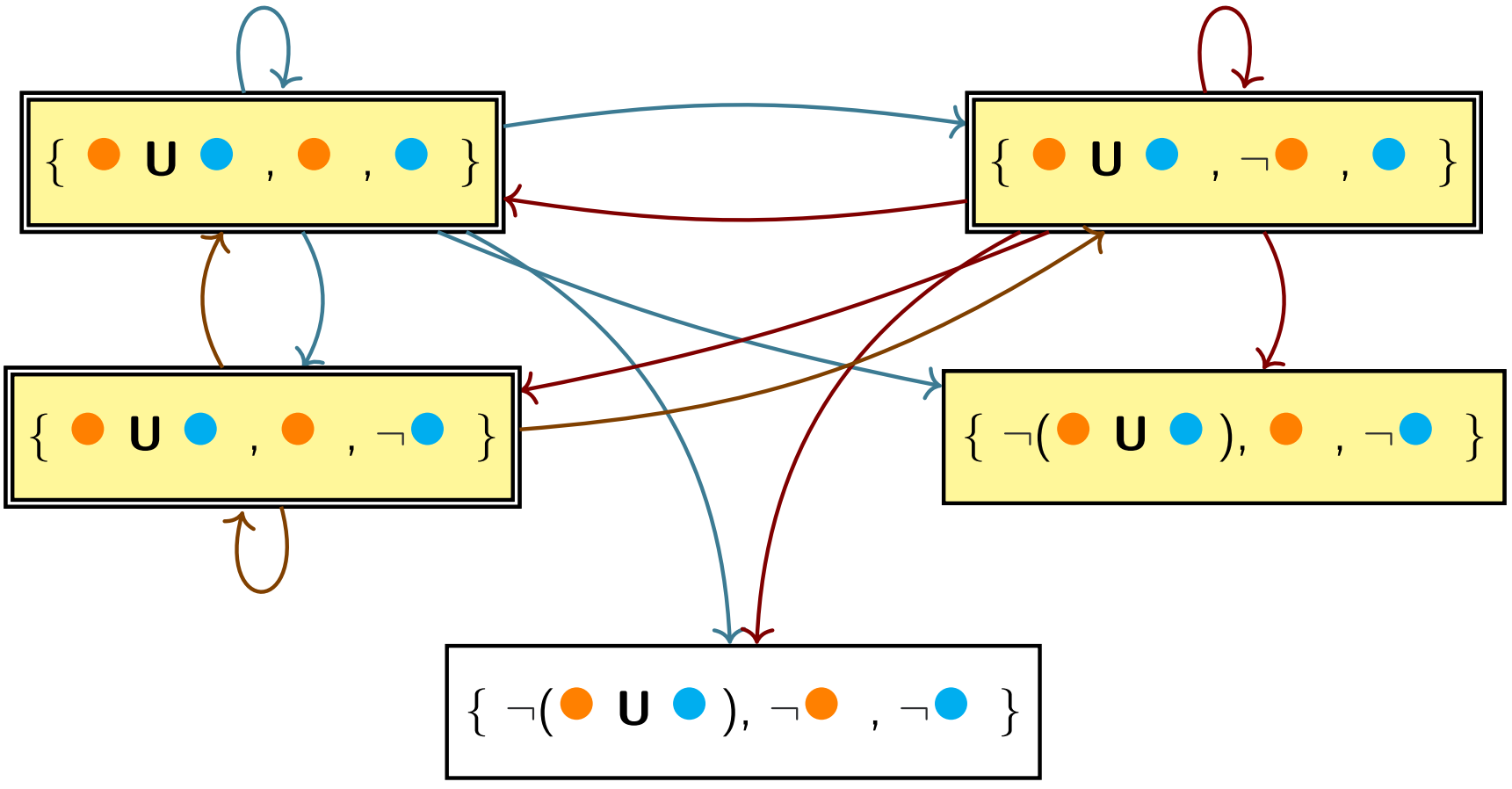

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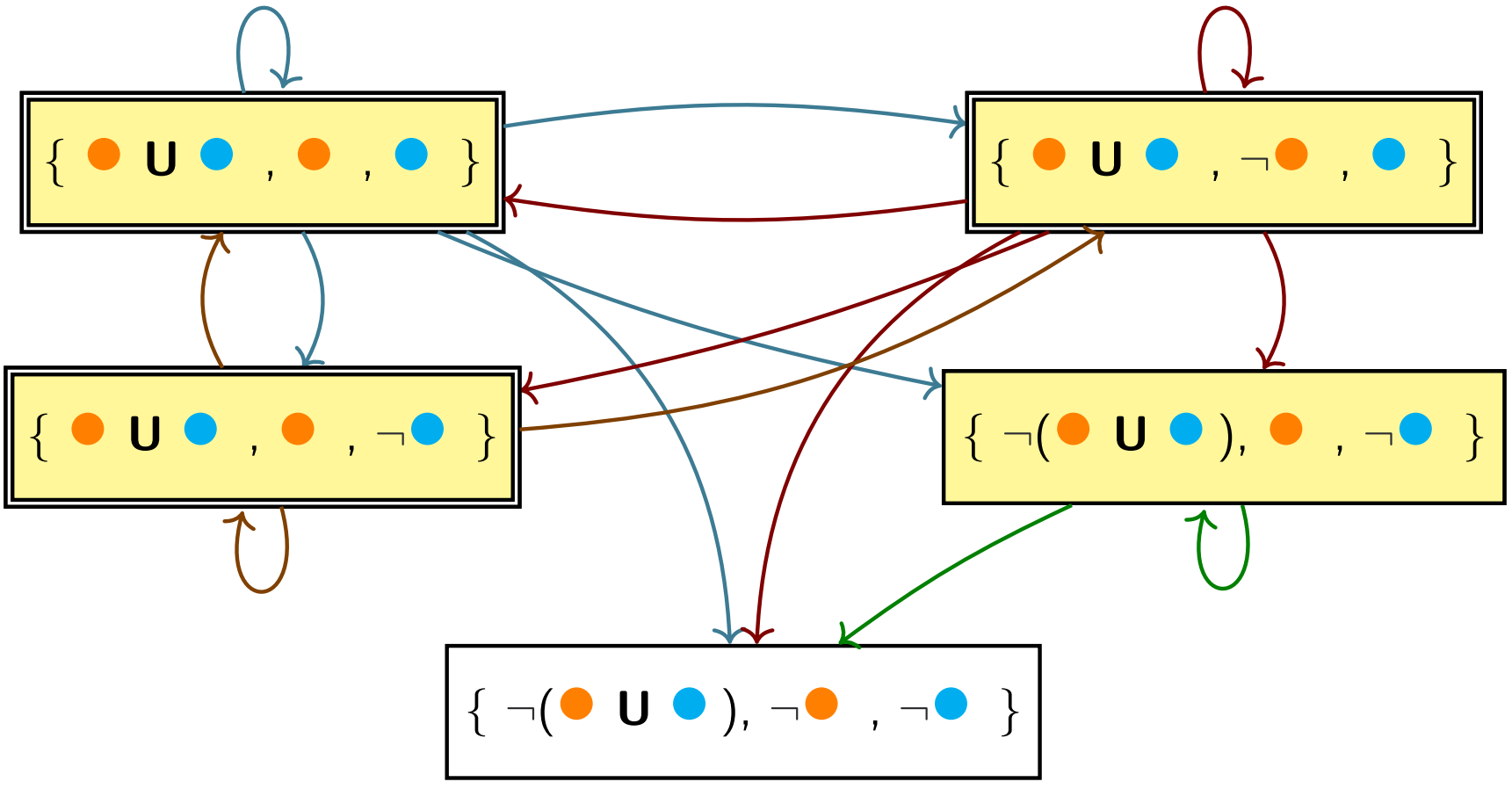

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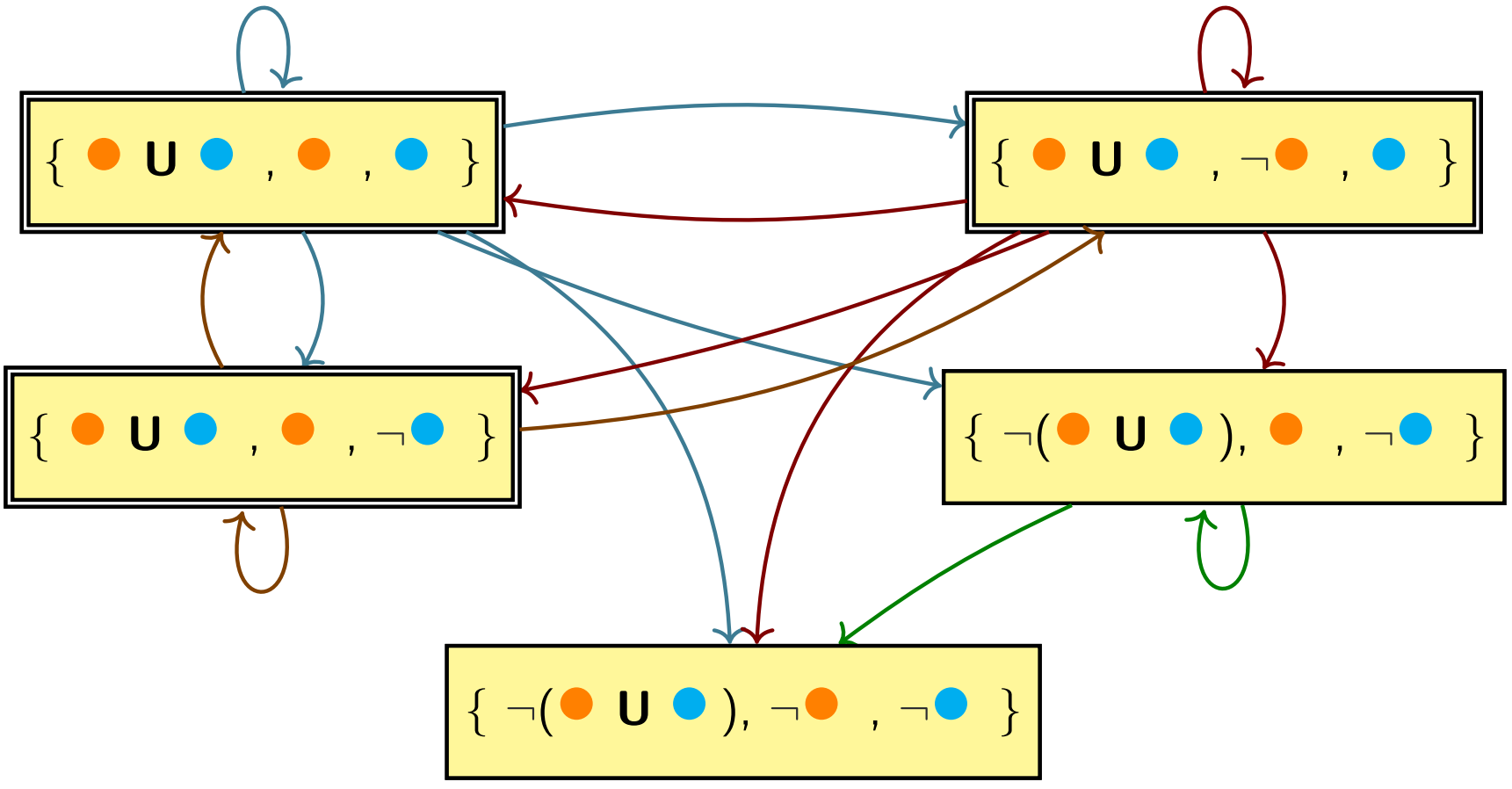

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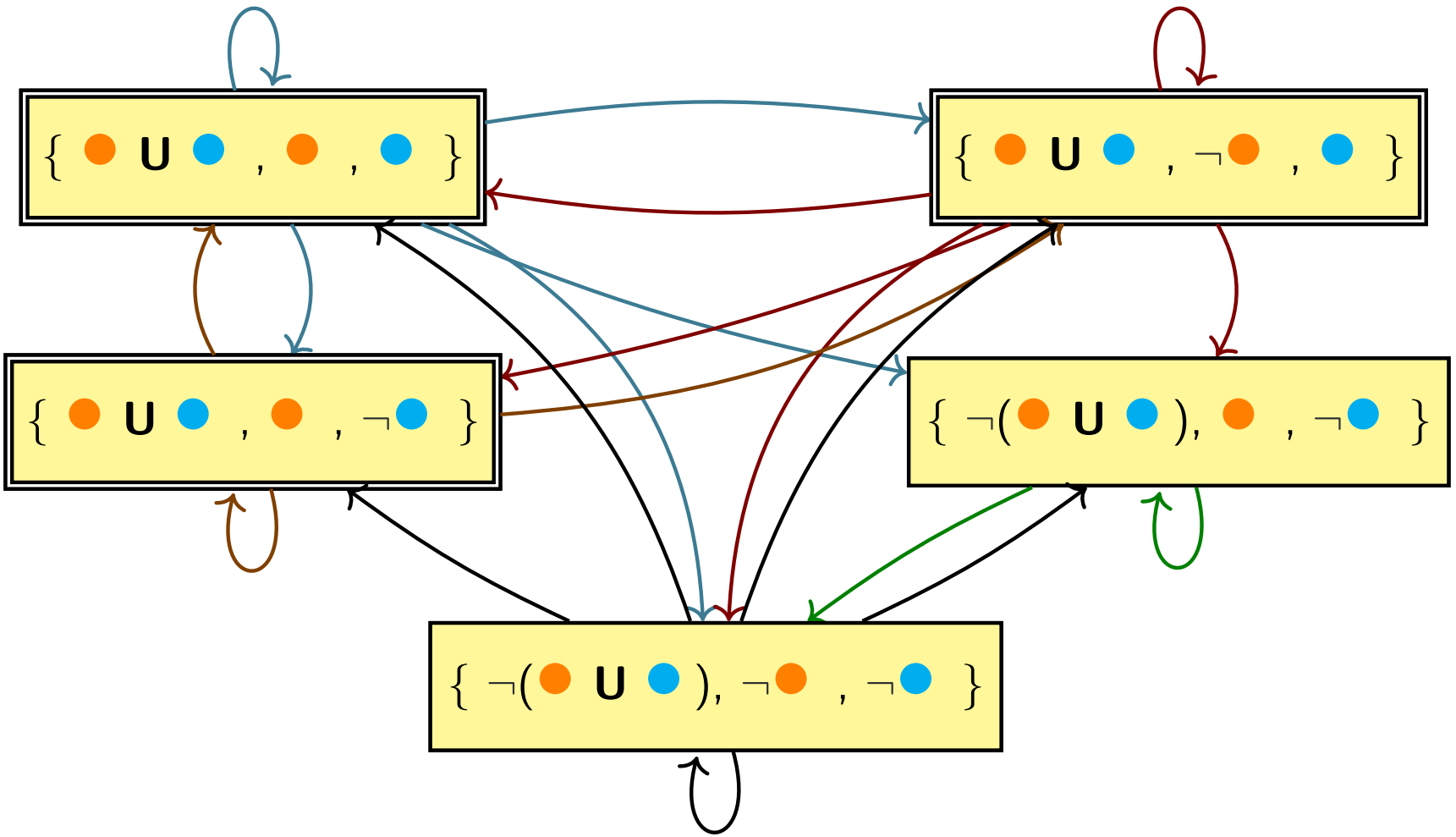

Eventuality Automaton

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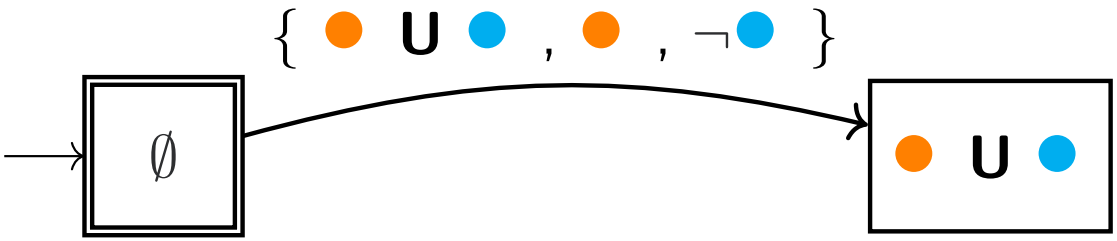

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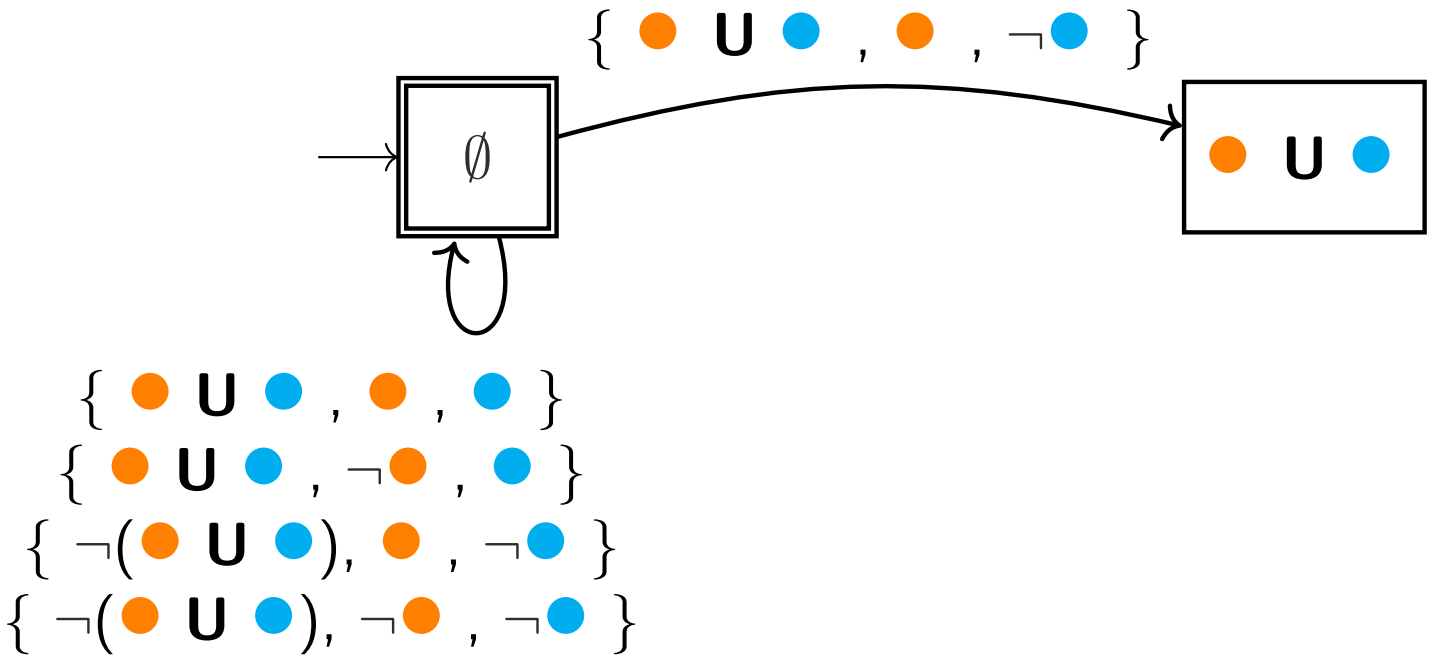

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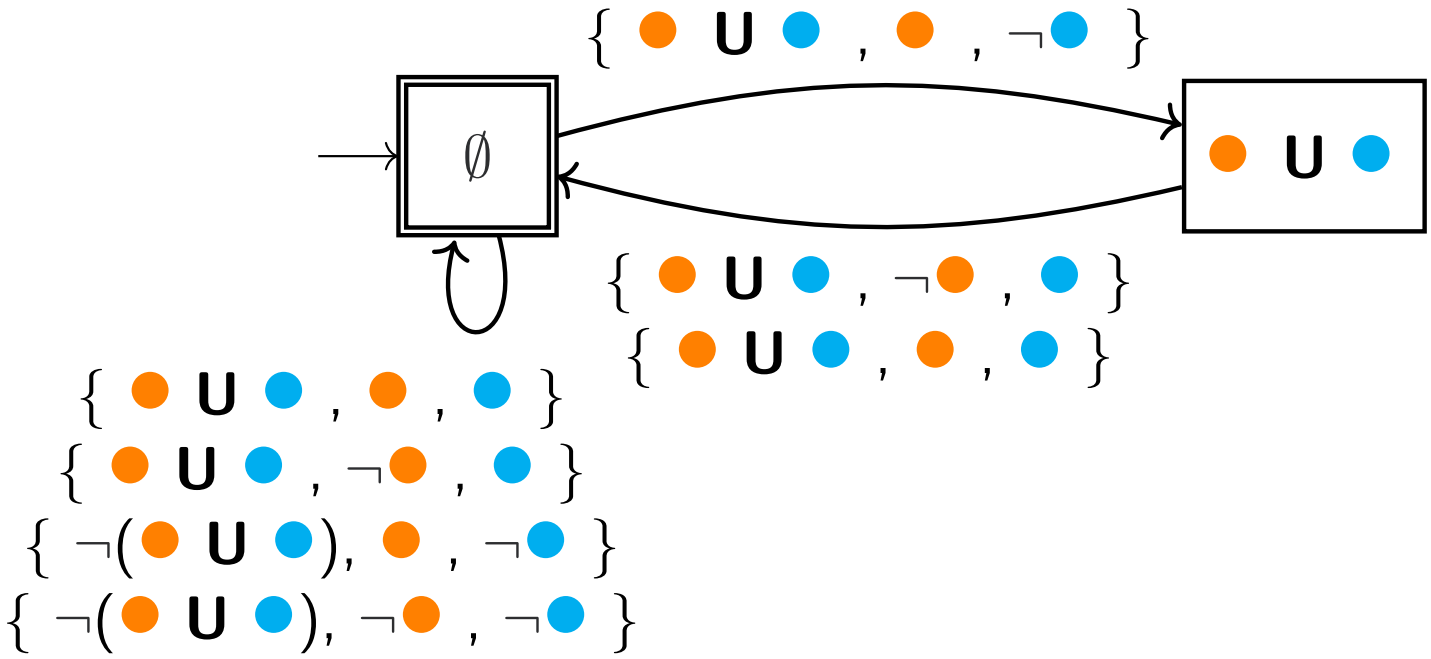

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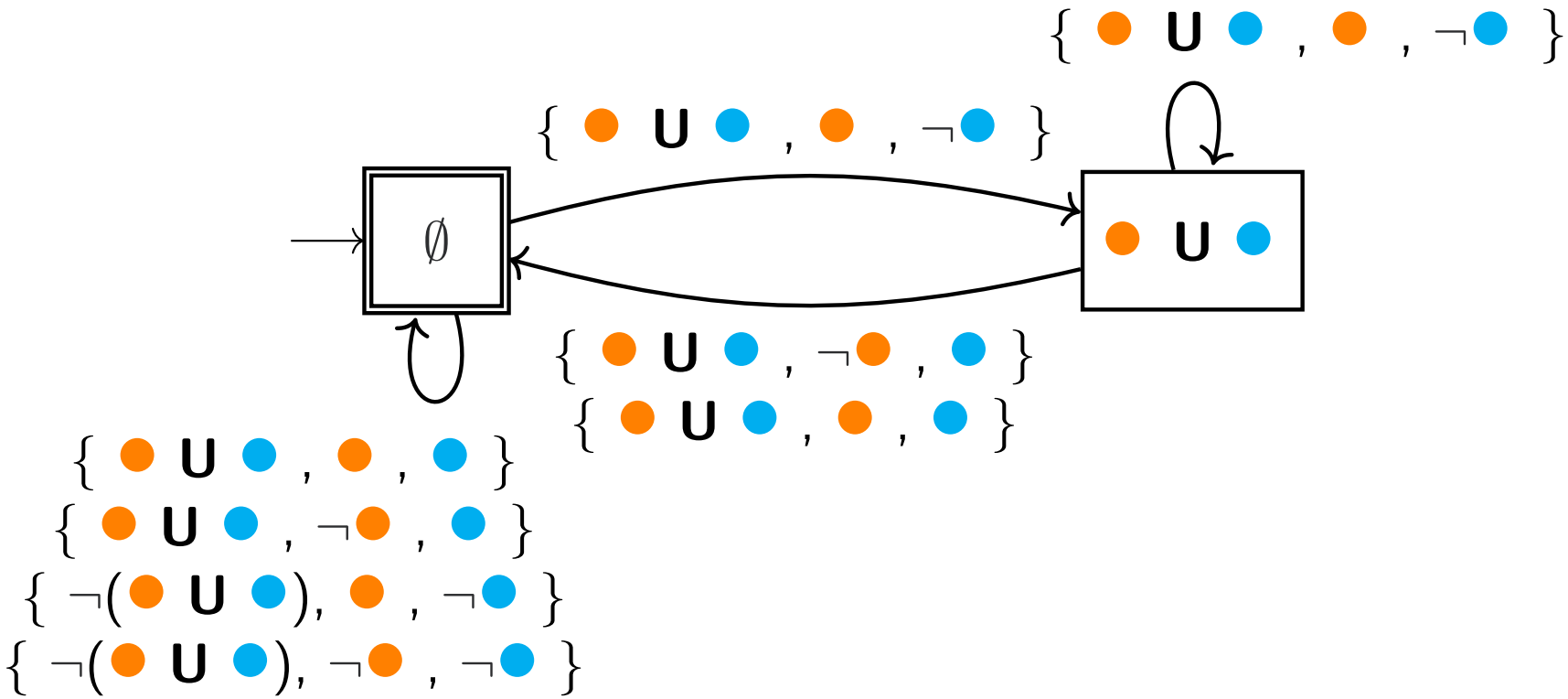

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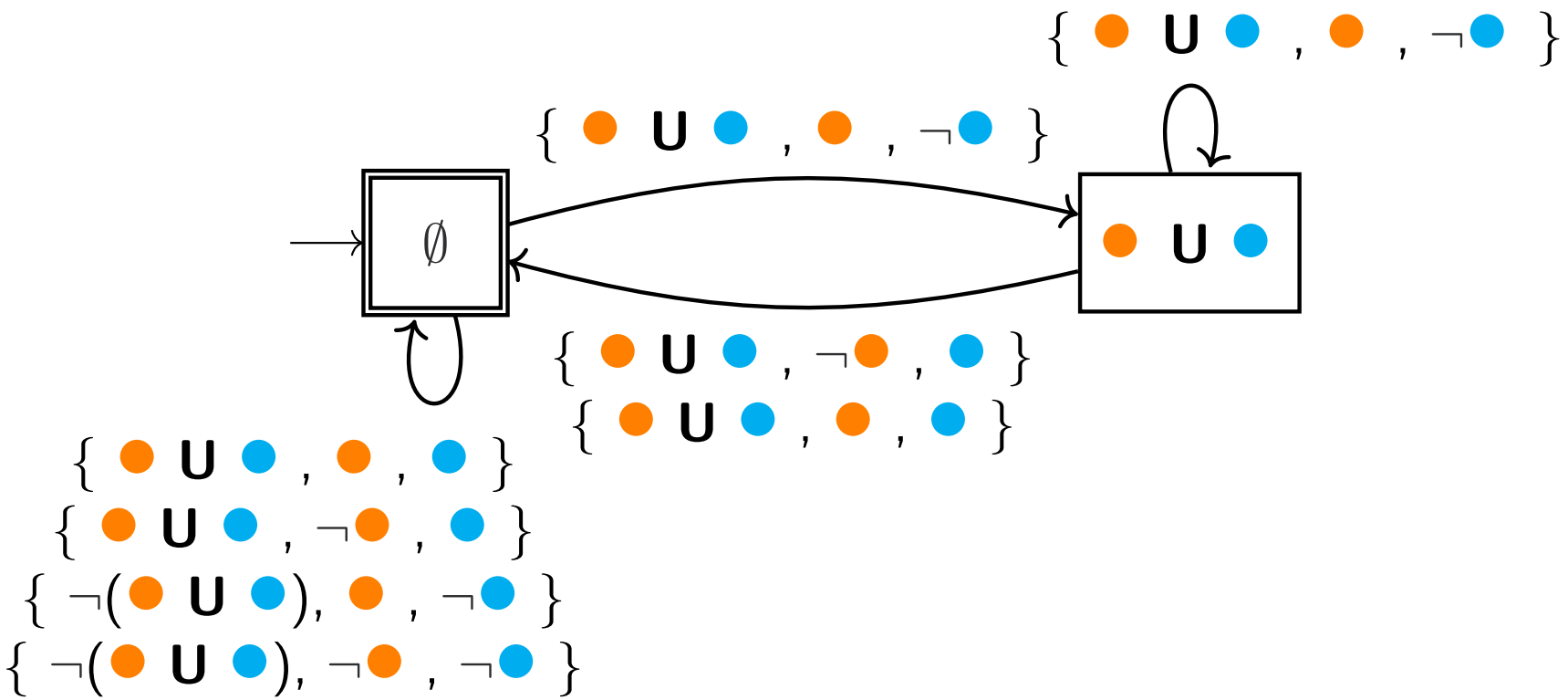
No other consistent edges!



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Our model automata A_M has just atomic propositions for actions, but our formula automaton $A^L_\Phi \times A^E_\Phi$ includes temporal propositions in the actions.

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Solution

After computing the product of local and eventuality automata, however, we can simply remove all negations and temporal propositions from the actions, leaving only atomic propositions behind.
Then we can compute the final product of our model with our negated formula as normal.

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Tons of overhead. Other methods are smarter (but even more complicated).

Bibliography

- Baier/Katoen: Principles of Model Checking, Section 5.2

Weekly Feedback

I would appreciate any comments/suggestions/requests you have on this week's lectures (and tutorials).

https://forms.office.com/r/sCi2xjavrS